Toward the Angular-Preservation Theorem: Reductions, a Non-Radiality Calculus, and the Green-Rank Frontier

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Abstract

This is a companion to "Keep the Angle: A Geometry-Preserving Basis in Spectral Embeddings" (Paper I). Paper I reduces the rank form of *Angular Preservation*—the conjecture that row-normalized spectral embeddings keep pairwise angular distances rank-faithful to geodesic distance—for intrinsic dimension $d \leq 3$, to two ingredients: a graph-to-continuum *rank transfer* and *Green-rank positivity* $\rho_G(M) = \rho_S(d_M(X,Y), -G_M(X,Y)) > 0$, where G_M is the zero-mean Green kernel. Here we move both toward theorems. We (i) delete the auxiliary hypotheses (H3) and atomlessness from Paper I's conditional results and reduce the graph-level rank transfer to *pairwise* anti-concentration alone; (ii) at a growing spectral cutoff *discharge* even that hypothesis—the graph angular ranking converges to $\rho_G(M)$ under calibration alone, so Angular Preservation (growing-window rank form) is *equivalent* to Green-rank positivity $\rho_G(M) > 0$—while at a fixed cutoff the operative *modulus* form (AC$'$), false as a bounded density, holds for every real-analytic metric and for every *smooth* metric under an explicit spectral nondegeneracy (ND); (iii) develop a *non-radiality calculus* that reduces Green-rank positivity to a single scale-invariant index of the Green kernel, via a master inequality $\rho_G \geq 1 - 12\,\phi\|R\|_\infty$ and its distributional refinement $\rho_G \geq 1 - 18(\phi^2 V)^{1/3}$; (iv) bound that index geometrically, identifying it with the *within-shell variance of the heat kernel*—which vanishes on two-point homogeneous spaces and yields a quantitative $\eta^{2/3}$ perturbative openness rate—leaving one named curvature-anisotropy amplitude as the sole non-explicit constant; and (v) carry the whole chain past the critical dimension via a fractional filter, proving the exact symmetric-space anchor ($\rho_G = 1$) in *every* dimension by heat-kernel subordination. A numerical counterexample hunt across four geometry families (the public repository) finds none. Green-rank positivity is thereby proved on an increasing hierarchy of explicit classes and reduced, in general, to one geometric quantity.

Disclosure and verification. Stated plainly: this work was developed with substantial assistance from an AI system (a large language model), and its internal checking was performed by fresh-context adversarial re-derivations run on the same class of system, supplemented by a further round of *AI-assisted* expert-style review (again a large language model, prompted adversarially). That process caught and corrected numerous errors in preparation—including a false $d = 4$ "coincidence" (now retracted) and gaps in the perturbative (§17) and subordination (§21) arguments (now repaired, the latter by grounding its analytic input in the published sharp heat-kernel estimate [15]). *No human expert has yet refereed this manuscript.* AI-assisted verification is not a substitute for human refereeing or machine-checked proof; the differential geometry of §§17–21 in particular is offered for expert scrutiny, while the elementary core (the master and distributional inequalities) has in addition been checked symbolically by computer algebra. Statements are made at the confidence their verification supports, and the author owns every claim.

1 Introduction

Paper I ("Keep the Angle" [1]) shows that the geometry a large graph Laplacian appears to lose to the commute-time collapse survives in the *angular* coordinate of the spectral embedding, and conjectures *Angular Preservation*: row-normalization keeps pairwise angular distances rank-faithful to geodesic distance, uniformly in the mode count. For $d \le 3$ the paper reduces the rank form to a chain,

$$\text{(graph angular ranking)} \xRightarrow{\text{rank transfer}} \text{(continuum angular ranking)} \xRightarrow{\text{Green-kernel rank limit}} \rho_G(M) > 0,$$

the last step being the Green-kernel rank-limit theorem, which sends the truncated commute angular ranking to the ranking of the zero-mean Green kernel G_M, and leaves open two things: the graph-level *rank transfer* (governed by a calibration error and an anti-concentration hypothesis) and *Green-rank positivity*, $\rho_G(M) := \rho_S\big(d_M(X,Y), -G_M(X,Y)\big) > 0$ for (X,Y) i.i.d. from the volume.

This companion collects and extends the reductions of both; at a growing spectral cutoff it closes the transfer outright (Part 11), leaving Green-rank positivity as the sole open link. Its contributions, by part:

- **Part 2 (reductions and the transfer hypothesis).** We show (H3), the injectivity hypothesis of Paper I's conditional theorem, is not independent—it follows from (H1)–(H2) under an explicit pinching condition with a quantitative angular gap; and that the atomlessness assumptions of the rank-limit results are automatic under a bounded pair density. We then prove the graph-level *rank transfer* needs neither bi-Lipschitz geometry nor a resolution cutoff—only calibration plus anti-concentration—and rederive the sharpened growing-window threshold $E_n\sqrt{A_m} \to 0$ by an independent route.
- **Part 7 (the anti-concentration hypothesis).** The transfer's anti-concentration hypothesis (AC), stated as a bounded density, is *false*—a nondegenerate saddle forces a logarithmic (van Hove) divergence, guaranteed on any closed surface of nonpositive Euler characteristic. We isolate the *modulus* form (AC′) the transfer proof actually consumes and prove it for every real-analytic metric, via a uniform Łojasiewicz sublevel estimate made uniform over the analytic family by o-minimality.
- **Part 11 (the unconditional growing window).** The transfer consumes only *pairwise* anti-concentration, so its uniform-in-anchor clause is deleted. At a growing spectral cutoff the hypothesis is *discharged*: along admissible schedules the graph angular ranking converges in probability to $\rho_G(M)$ with nothing assumed beyond calibration, so the growing-window rank form of Angular Preservation is *equivalent* to $\rho_G(M) > 0$—the conjecture *is* the geometry. At a fixed cutoff, atomlessness of the angular pair law—hence (AC′)—holds for every *smooth* metric under an explicit spectral nondegeneracy (ND), automatic in the analytic category; whether (ND) can fail for some smooth metric is the sole residual.

- **Part 12 (a non-radiality calculus).** We reduce Green-rank positivity to a single quantity. A *master inequality* $\rho_G \geq 1 - 12\,\phi_\Psi \|R\|_\infty$ controls the rank deficit by the sup-norm non-radiality of $-G$ against its best monotone-radial profile; a *distributional refinement* $\rho_G \geq 1 - 18(\phi_\Psi^2 V)^{1/3}$ replaces the sup by the L^2 non-radiality V, with the index $\phi_\Psi^2 V$ dimensionless (scale-invariant)—as a bound on a rank correlation must be. Two-point homogeneous spaces are the exact ($R \equiv 0$) anchor.

- **Part 17 (the geometric bound).** We identify the L^2 non-radiality with the *within-distance-shell variance of the Green kernel*, and via the heat representation with the integrated shell-variance of the heat kernel: it vanishes on two-point homogeneous spaces, is finite for $d < 8$ (the shell-centered part scaling as $t^{1-d/4}$; the operative $d \leq 3$ comes from the Hilbert–Schmidt/rank-limit step, *not* the shell-variance—a point corrected from an earlier over-claim), has an explicit spectral-gap tail, and is $O(\|g - g_0\|_{C^2}^2)$ near any symmetric metric—so $\rho_G \geq 1 - C\eta^{2/3} \to 1$, a *quantitative* form of Paper I's qualitative openness. One named curvature-anisotropy amplitude remains the sole non-explicit constant.

- **Part 21 (past the critical dimension).** For $d \geq 4$ the commute Green kernel leaves L^2; a fractional filter $\mu_k^{-s/2}$, $s > d/4$, restores the whole chain, and heat-kernel subordination proves the exact anchor—fractional Green-rank $= 1$ on all rank-one symmetric spaces—in *every* dimension.

The upshot (§26): Green-rank positivity is proved exactly on two-point homogeneous spaces (all dimensions), quantitatively on their C^2-neighborhoods, and on the explicit near-radial / L^2-near-radial classes; in general it is reduced to positivity of a single geometric coefficient, with a numerical counterexample hunt (four families) finding none. Notation follows Paper I and its supplement throughout; the phrase "companion note" in the assembled parts below refers to the other parts of this paper.

2 Reductions and the transfer hypothesis

Purpose. This note records four self-contained results that move the rank form of Angular Preservation (Paper I, Conjecture on Angular Preservation) toward a theorem. Notation follows Paper I and its supplement: M is a closed Riemannian d-manifold with volume normalized to 1; $F = (\varphi_k/\sqrt{\mu_k})_{k\leq m}$ (or its spectral-cutoff form F_Λ) is the commute-weighted eigenmap, $R = \|F\|$, $N = F/R$; (H1)–(H3), the calibration error E_n, the radius band $[\rho^-, \rho^+]$, the Green-rank coefficient $\rho_G(M) = \rho_S\big(d_M(X,Y), -G_M(X,Y)\big)$, the diagonal scale A_Λ, and the Green-kernel rank-limit theorem are as in Paper I. Facts imported as standard are listed in §5 with sources.

The upshot (§26): for $d \leq 3$, the rank form of Angular Preservation reduces to

$$\underbrace{\rho_G(M) > 0}_{\text{geometric}} + \underbrace{\text{(AC)}}_{\text{anti-concentration}} + \underbrace{E_n \to 0 \;\; \text{(fixed } m) \;/\; E_n\sqrt{A_m} \to 0 \;\; \text{(growing } m)}_{\text{calibration}},$$

with Theorem 6 proving the first item on a C^2-open set of metrics, Theorem 8 proving the transfer given the second and third, and Lemmas 2 and 4 deleting hypotheses ((H3); atomlessness) from the existing results.

3 (H3) is not an independent hypothesis

Recall the setting of Paper I's conditional theorem: (H1) is L_0-bi-Lipschitzness of F on geodesic r_0-balls; (H2) is $\rho^- \le R \le \rho^+$ with R globally Λ_0-Lipschitz and $\Lambda_0 < 1/L_0$; (H3) is injectivity of N on M. The key observation is an exact identity for radial alignment.

Lemma 1 (Radial-alignment identity). *If $N(x) = N(y)$ then $F(x) = tF(y)$ with $t = R(x)/R(y) > 0$, and*

$$\|F(x) - F(y)\| = |t-1|\, R(y) = |R(x) - R(y)|.$$

Proof. $N(x) = N(y)$ means $F(x)/R(x) = F(y)/R(y)$, i.e. $F(x) = tF(y)$ with $t = R(x)/R(y)$. Then $\|F(x) - F(y)\| = |t-1|\|F(y)\| = |tR(y) - R(y)| = |R(x) - R(y)|$. □

Lemma 2 ((H3) from (H1)–(H2)). *Assume (H1) on r_0-balls and (H2), and set*

$$c_F := \min\{\|F(x) - F(y)\| :\ d_M(x,y) \ge r_0\}, \qquad \delta := \min\big(\rho^+ - \rho^-,\ \Lambda_0 \operatorname{diam} M\big).$$

Then:

(a) N is injective on every r_0-ball (no hypothesis beyond (H1)–(H2));

(b) if $c_F > \delta$, then (H3) holds on all of M, and quantitatively

$$c_0 := \min\{\|N(x) - N(y)\| :\ d_M(x,y) \ge r_0\} \ \ge\ \frac{\sqrt{c_F^2 - \delta^2}}{\rho^+} \ >\ 0;$$

(c) if (H1) holds globally ($r_0 \ge \operatorname{diam} M$), (H3) holds with no further condition; if R is constant, (H3) holds if and only if $c_F > 0$, i.e. iff F does not identify any pair at distance $\ge r_0$.

Proof. (a) Suppose $N(x) = N(y)$ with $0 < d := d_M(x,y) \le r_0$. By Lemma 1 and (H2), $\|F(x) - F(y)\| = |R(x) - R(y)| \le \Lambda_0 d$; by (H1), $\|F(x) - F(y)\| \ge d/L_0$. Hence $d/L_0 \le \Lambda_0 d$, contradicting $\Lambda_0 < 1/L_0$.

(b) Suppose $N(x) = N(y)$ with $d \ge r_0$. By Lemma 1, $\|F(x) - F(y)\| = |R(x) - R(y)| \le \delta$ (the two bounds $\rho^+ - \rho^-$ and $\Lambda_0 d \le \Lambda_0 \operatorname{diam} M$ both apply). But $\|F(x) - F(y)\| \ge c_F > \delta$: contradiction, so no far pair aligns; together with (a), N is injective. For the quantitative bound, apply Paper I's normalization identity to a far pair:

$$\|N(x) - N(y)\|^2 = \frac{\|F(x) - F(y)\|^2 - \big(R(x) - R(y)\big)^2}{R(x)R(y)} \ \ge\ \frac{c_F^2 - \delta^2}{(\rho^+)^2},$$

using $\|F(x) - F(y)\| \ge c_F$, $|R(x) - R(y)| \le \delta$, and $R(x)R(y) \le (\rho^+)^2$.

(c) If $r_0 \ge \operatorname{diam} M$, every pair falls under case (a). If R is constant then $\rho^+ = \rho^-$ and $\delta = 0$, so (b) applies whenever $c_F > 0$; conversely if $c_F = 0$ some far pair has $F(x) = F(y)$, hence $N(x) = N(y)$, and (H3) fails. Compactness makes c_F an attained minimum, so $c_F > 0$ is equivalent to "F identifies no pair at distance $\ge r_0$". □

Remark 3. Consequences for Paper I. (i) The global corollary (Cor. "Global conditional form") can replace hypothesis (H3) by the checkable pinching condition $c_F > \delta$, and its nonconstructive constant c_0 becomes the explicit $\sqrt{c_F^2 - \delta^2}/\rho^+$, making A_g fully quantitative. (ii) Portegies-type mode counts give injectivity of F, hence $c_F > 0$ by compactness; the remaining condition is the pinching $c_F > \delta$. (iii) The homogeneous-space remark of Paper I is the case $\rho^+ = \rho^-$ of (c). (iv) A distance-resolved refinement holds with the same proof: alignment at distance d is impossible whenever $\min\{\|F(x) - F(y)\| : d_M(x,y) = d\} > \min(\rho^+ - \rho^-, \Lambda_0 d)$.

4 Atomlessness is automatic

Paper I's rank-limit results (the flat-torus proposition and the Green-kernel rank-limit theorem) assume that the laws of $d_M(X,Y)$ and $G(X,Y)$ are atomless. Both assumptions can be deleted.

We use twice the standard level-set lemma: *if $h \in C^1$ on an open subset of $\mathbb{R}^d$ (or a manifold chart), then $\nabla h = 0$ almost everywhere on $\{h = c\}$* (see §5). Iterating: if $f \in C^2$ and $E = \{f = c\}$ has positive measure, then a.e. on E we have $\nabla f = 0$; since $E \subseteq \{\partial_i f = 0\}$ up to a null set for each i, a second application (to $h = \partial_i f$) gives $\nabla \partial_i f = 0$ a.e. on E; hence the coordinate Hessian vanishes a.e. on E, and therefore so does the Laplace–Beltrami operator, $\Delta_g f = g^{ij}(\partial_i \partial_j f - \Gamma^k_{ij} \partial_k f) = 0$ a.e. on E.

Lemma 4 (Automatic atomlessness). *Let M be a closed smooth Riemannian manifold (any dimension $d \geq 1$, volume normalized to 1), G its zero-mean Green kernel, and let (X,Y) have a law with bounded density with respect to* $\mathrm{vol} \otimes \mathrm{vol}$. *Then the laws of $d_M(X,Y)$ and of $G(X,Y)$ are atomless.*

Proof. By the bounded-density assumption it suffices to show that for each fixed x the sets $\{y : d_M(x,y) = t\}$ and $\{y : G(x,y) = c\}$ are vol-null; Fubini then gives $\mathbb{P}(d_M(X,Y) = t) = \mathbb{P}(G(X,Y) = c) = 0$.

Distance. Fix x. The function $d_x := d_M(x,\cdot)$ is 1-Lipschitz, and on $M \setminus (\{x\} \cup \mathrm{Cut}(x))$ it is smooth with $|\nabla d_x| = 1$ (the eikonal identity); the cut locus $\mathrm{Cut}(x)$ is closed and vol-null (imported, §5). By the level-set lemma applied to the Lipschitz function d_x, $\nabla d_x = 0$ a.e. on $\{d_x = t\}$. Off the null set $\{x\} \cup \mathrm{Cut}(x)$ this contradicts $|\nabla d_x| = 1$, so $\mathrm{vol}\{d_x = t\} = 0$.

Green kernel. Fix y. By elliptic regularity $G(\cdot,y)$ is smooth on $M \setminus \{y\}$ and satisfies, with the positive Laplacian and unit volume, $\Delta\, G(\cdot,y) = -1$ there. If $\{x : G(x,y) = c\}$ had positive measure, the iterated level-set lemma above would give $\Delta G(\cdot,y) = 0$ on a positive-measure subset, contradicting $\Delta G(\cdot,y) \equiv -1$. So each slice is null. □

Remark 5. (i) Consequently the flat-torus rank proposition and the Green-kernel rank-limit theorem of Paper I hold under the single hypothesis of a bounded pair density: their atomlessness assumptions are theorems, not hypotheses. (ii) The argument does not apply to the *truncated* kernels G_Λ (there $\Delta_x G_\Lambda(\cdot,y) = E_\Lambda(\cdot,y)$ is not a nonzero constant), but the limit-law atomlessness is all those proofs use. (iii) The argument is dimension-independent; it does not use $d \leq 3$.

5 Green-rank positivity is an open condition

Theorem 6 (Perturbative openness of Green-rank positivity). *Let M_0 be a closed smooth manifold of dimension $d \leq 3$ and let g_0 be a smooth Riemannian metric on M_0, volume normalized to 1. Then the map $g \mapsto \rho_G(g) := \rho_S\big(d_g(X,Y), -G_g(X,Y)\big)$ (with (X,Y) i.i.d. from normalized vol_g) is continuous at g_0 with respect to C^2 convergence of metrics, and likewise for the Kendall version τ_G. Consequently:*

(a) if $\rho_G(g_0) > 0$, then $\rho_G > 0$ on a C^2-neighborhood of g_0;

(b) in particular, every two-point homogeneous metric of dimension ≤ 3 ($S^2, S^3, \mathbb{RP}^2, \mathbb{RP}^3$; and every metric on S^1) has a C^2-neighborhood on which $\rho_G > 0$;

(c) *combined with the Green-kernel rank-limit theorem of Paper I (whose atomlessness hypotheses are deleted by Lemma 4), for every g in such a neighborhood there exist $\Lambda_0(g) < \infty$ and $\rho_0(g) > 0$ such that the truncated commute angular ranking satisfies $\rho_S\big(d_g(X,Y), \|N_\Lambda(X) - N_\Lambda(Y)\|\big) \geq \rho_0(g)$ for every multiplet-closed $\Lambda \geq \Lambda_0(g)$: the continuum rank form of Angular Preservation, uniform in the mode count, holds on a C^2-open set of metrics.*

Proof. Throughout, $g \to g_0$ in C^2; after a global rescaling (continuous in g) we may keep $\mathrm{vol}_g(M) = 1$. Write $\mathrm{vol}_0 = \mathrm{vol}_{g_0}$ and $\theta_g = d\,\mathrm{vol}_g / d\,\mathrm{vol}_0 \to 1$ in C^1.

Step 1: distances. If $(1+\eta)^{-1} g_0 \leq g \leq (1+\eta) g_0$ as quadratic forms (which holds with $\eta = \eta(g) \to 0$), then $(1+\eta)^{-1/2} d_{g_0} \leq d_g \leq (1+\eta)^{1/2} d_{g_0}$, so $d_g \to d_{g_0}$ uniformly on $M \times M$.

Step 2: uniform spectral tails. The Dirichlet forms $\mathcal{E}_g(u) = \int |\nabla u|_g^2 \, d\,\mathrm{vol}_g$ and the L^2 norms are uniformly comparable with constants $(1+\eta)^{\pm c(d)}$, so by min–max $\mu_k(g) \geq (1+\eta)^{-c(d)} \mu_k(g_0)$ for all k. Since $d \leq 3$ gives $\sum_k \mu_k(g_0)^{-2} < \infty$, the tails $\sum_{k>K} \mu_k(g)^{-2} \leq (1+\eta)^{2c} \sum_{k>K} \mu_k(g_0)^{-2}$ are small uniformly over the neighborhood once K is large.

Step 3: heads via resolvents. Let $J_g : L^2(\mathrm{vol}_g) \to L^2(\mathrm{vol}_0)$, $J_g u = \theta_g^{1/2} u$, a unitary; $A_g := J_g \Delta_g J_g^{-1}$ is self-adjoint on the fixed space $L^2(\mathrm{vol}_0)$ with the same spectrum, and its quadratic form $a_g(v) = \mathcal{E}_g(\theta_g^{-1/2} v)$ has coefficients converging uniformly to those of a_{g_0} (this uses $\theta_g \to 1$ in C^1, hence $g \to g_0$ in C^2 suffices; C^1 is what the computation needs). Uniformly comparable closed forms converging in this sense yield norm resolvent convergence $A_g \to A_{g_0}$ (imported, §5). Fix K as in Step 2 with $\mu_K(g_0) < \mu_{K+1}(g_0)$, and a contour $\Gamma \subset \mathbb{C} \setminus \mathrm{spec}(A_{g_0})$ enclosing $\{\mu_1(g_0), \ldots, \mu_K(g_0)\}$ but not 0 nor $\mu_{K+1}(g_0)$. For g near g_0, Γ still separates the same eigenvalue count (eigenvalue continuity from Step 2's two-sided comparison), and

$$\mathrm{Head}_g := \frac{1}{2\pi i} \oint_\Gamma z^{-1} (z - A_g)^{-1} \, dz = \sum_{k \leq K} \mu_k(g)^{-1} \, P_k(g)$$

(the residue of $z^{-1}(z - A_g)^{-1}$ at an eigenvalue μ is μ^{-1} times its spectral projector; 0 lies outside Γ, so the constant mode does not contribute). Norm resolvent convergence gives $\mathrm{Head}_g \to \mathrm{Head}_{g_0}$ in operator norm; both sides have rank K, so the difference has rank $\leq 2K$ and $\|T\|_{\mathrm{HS}} \leq \sqrt{\mathrm{rank}\, T}\, \|T\|_{\mathrm{op}}$ upgrades this to Hilbert–Schmidt convergence, i.e. $L^2(M \times M)$ convergence of the head kernels. Undoing the unitary multiplies kernels by $\theta_g^{-1/2}(x) \theta_g^{-1/2}(y) \to 1$ uniformly. Combining with Step 2, $G_g \to G_{g_0}$ in $L^2(M \times M, \mathrm{vol}_0 \otimes \mathrm{vol}_0)$.

Step 4: joint weak convergence of the pair statistics. For bounded continuous h,

$$\mathbb{E}_g \, h\big(d_g(X,Y), G_g(X,Y)\big) = \int h\big(d_g, G_g\big) \, \theta_g(x) \theta_g(y) \, d\,\mathrm{vol}_0 \, d\,\mathrm{vol}_0 \, .$$

The densities $\to 1$ uniformly; $d_g \to d_{g_0}$ uniformly; $G_g \to G_{g_0}$ in $L^2(\mathrm{vol}_0^{\otimes 2})$, hence in measure; so $h(d_g, G_g) \to h(d_{g_0}, G_{g_0})$ in measure and, being bounded, in $L^1(\mathrm{vol}_0^{\otimes 2})$. Thus $\big(d_g(X,Y), G_g(X,Y)\big) \Rightarrow \big(d_{g_0}(X,Y), G_{g_0}(X,Y)\big)$.

Step 5: continuity of the rank functionals. By Lemma 4 the marginal laws of $d_g(X,Y)$ and $G_g(X,Y)$ are atomless for *every* smooth g, so $\rho_G(g)$ is the honest grade correlation $\rho_G(g) = 12\,\mathbb{E}\big[F_{d_g}(d_g)\, F_{-G_g}(-G_g)\big] - 3$. Marginal weak convergence to atomless limits gives uniform convergence of the CDFs (Pólya); replacing F_{d_g}, F_{-G_g} by $F_{d_{g_0}}, F_{-G_{g_0}}$ costs $o(1)$, and the remaining expectation converges by Step 4 since $(u,v) \mapsto F_{d_{g_0}}(u) F_{-G_{g_0}}(-v)$ is bounded and continuous. Hence $\rho_G(g) \to \rho_G(g_0)$. The Kendall functional, written on two independent pairs, converges by the same weak-convergence-plus-null-discontinuity argument used in Paper I's circle theorem.

Conclusions. (a) is immediate. (b): two-point homogeneity gives $\rho_G(g_0) = 1$ by Paper I's corollary; on S^1 every smooth metric is isometric to a round circle (arc-length parametrization), so $\rho_G \equiv 1$ there outright. (c) combines (a) with the Green-kernel rank-limit theorem, whose only remaining hypothesis (bounded pair density) holds for i.i.d. volume sampling. □

Remark 7. The neighborhood in Theorem 6 is not effective: Step 3 uses qualitative norm resolvent convergence. An effective radius would follow from quantitative resolvent perturbation bounds in terms of $\|g - g_0\|_{C^1}$ and the spectral gap at $\mu_K(g_0)$; this is routine but unpleasant, and is left as the quantitative frontier.

6 An anti-concentration transfer theorem (graph level)

The conditional theorem of Paper I transfers the *metric* through (H1)–(H2). For the *rank* clause none of that is needed: calibration plus anti-concentration suffice, with no resolution cutoff θ_n and no bi-Lipschitz hypothesis.

Fix a multiplet-closed cutoff Λ and write $V(x, y) = \|N_\Lambda(x) - N_\Lambda(y)\|$ for the continuum angular distance and $D(x, y) = d_M(x, y)$. For an i.i.d. sample $x_1, \ldots, x_n$ from a bounded, bounded-below density, let $\widehat{V}_{ij}$ be the graph angular distances and suppose the calibration event of Paper I's transfer lemma holds: $\max_i \|\hat{q}_i - N_\Lambda(x_i)\| \le 2E_n/\rho_\Lambda^-$, hence

$$|\widehat{V}_{ij} - V(x_i, x_j)| \;\le\; \bar{\varepsilon}_n := \frac{4E_n}{\rho_\Lambda^-} \qquad \text{for } \textit{every} \text{ pair } i \ne j \tag{1}$$

(the per-node bound holds at every node; no distance restriction enters).

(AC) Anti-concentration. $\kappa_\Lambda := \sup\limits_{x \in M} \big\|\text{density of } V(x, Y)\big\|_\infty < \infty$, where Y is drawn from the sampling density.

Theorem 8 (Rank transfer without bi-Lipschitz hypotheses)**.** *Let $d \le 3$ and suppose $\rho_G(M) > 0$ (e.g. M in the open set of Theorem 6), so that by the Green-kernel rank-limit theorem there are Λ_0, $\rho_\star > 0$, $\tau_\star > 0$ with $\rho_S(D, V) \ge \rho_\star$ and $\tau_K(D, V) \ge \tau_\star$ for every multiplet-closed $\Lambda \ge \Lambda_0$. Fix such a Λ and assume (AC) and the calibration* (1) *with $E_n \to 0$. Then, with probability tending to one, the* empirical *rank correlations of the graph angular distances against the manifold distances over all sampled pairs satisfy*

$$\widehat{\tau}_K\big(\widehat{V}, D\big) \;\ge\; \tau_\star - C\,\kappa_\Lambda \bar{\varepsilon}_n - o(1), \qquad \widehat{\rho}_S\big(\widehat{V}, D\big) \;\ge\; \rho_\star - C\,\kappa_\Lambda \bar{\varepsilon}_n - o(1),$$

with an absolute constant C; in particular both are eventually bounded below by $\tau_\star/2$ and $\rho_\star/2$ respectively.

Remark 9 ((AC) is to be read as the modulus form (AC′))*.* The bounded-density hypothesis (AC) as displayed is *false* in general—a nondegenerate saddle of $V(x, \cdot)$ forces a van Hove logarithmic divergence of κ_Λ (§7, guaranteed on any closed surface of Euler characteristic ≤ 0). The proof, however, uses only the weaker *modulus* form

$$\text{(AC}'\text{)} \qquad \sup_{x \in M} \sup_{t} \; \mathbb{P}_Y\big(|V(x, Y) - t| \le \varepsilon\big) \;\le\; \omega(\varepsilon), \quad \omega(0^+) = 0,$$

and the conclusion holds verbatim with $\kappa_\Lambda \bar{\varepsilon}_n$ replaced by $\omega(2\bar{\varepsilon}_n)$, hence under $\omega(2\bar{\varepsilon}_n) \to 0$. §7 establishes this substitution and proves (AC′) for every real-analytic metric; throughout, (AC) is to be understood as (AC′).

Proof. Write $N_p = \binom{n}{2}$ for the number of pairs, V_P for the continuum value of pair P and $\widehat{V}_P$ for the graph value. Ties in V are harmless: a tied pair $|V_P - V_Q| = 0 \le 2\bar{\varepsilon}_n$ already lies inside the flip event that Step 1 counts, provided both rankings use one fixed tie-breaking (midrank) convention; and the empirical-to-population step, read in the midrank convention, converges because the limit (Green) law is atomless (Lemma 4).

Step 1: perturbation flips few comparisons. For a pair-of-pairs $\{P, Q\}$, the comparison $\operatorname{sgn}(\widehat{V}_P - \widehat{V}_Q)$ can differ from $\operatorname{sgn}(V_P - V_Q)$ only if $|V_P - V_Q| \le 2\bar{\varepsilon}_n$, by (1). Let f_n be the fraction of pairs-of-pairs with $|V_P - V_Q| \le 2\bar{\varepsilon}_n$. Then

$$\left|\widehat{\tau}_K(\widehat{V}, D) - \widehat{\tau}_K(V, D)\right| \le 2f_n.$$

For Spearman: the V-ranking and the $\widehat{V}$-ranking of the N_p pairs differ by a permutation whose Kendall distance is at most $f_n\binom{N_p}{2}$; by the Diaconis–Graham inequality the footrule (total rank displacement) is at most twice the Kendall distance, and since one unit of rank displacement changes $\sum_i d_i^2$ in the Spearman formula $\widehat{\rho}_S = 1 - 6\sum d_i^2/(N_p^3 - N_p)$ by at most $2N_p$, we get

$$\left|\widehat{\rho}_S(\widehat{V}, D) - \widehat{\rho}_S(V, D)\right| \;\le\; \frac{6 \cdot 2N_p \cdot 2f_n\binom{N_p}{2}}{N_p^3 - N_p} \;\le\; 12f_n + O(1/N_p).$$

Step 2: the flip fraction is small in expectation. Take a pair-of-pairs $\{P, Q\}$. If P, Q are vertex-disjoint, condition on V_Q: by (AC), integrating the conditional density of V_P given one endpoint (then averaging over it) bounds the conditional density of V_P by κ_Λ, so $\mathbb{P}(|V_P - V_Q| \le 2\bar{\varepsilon}_n) \le 4\kappa_\Lambda\bar{\varepsilon}_n$. If $P = (x_i, x_j)$ and $Q = (x_i, x_k)$ share the anchor x_i, condition on x_i and V_Q: given x_i, the value $V_P = V(x_i, x_j)$ with x_j independent has density $\le \kappa_\Lambda$ by (AC) (this is exactly why (AC) is stated uniformly in the anchor), so the same bound holds. Hence $\mathbb{E}f_n \le 4\kappa_\Lambda\bar{\varepsilon}_n$, and by Markov $f_n \le \kappa_\Lambda\bar{\varepsilon}_n^{1/2}\cdot$(const), say, with probability tending to one; any rate with $f_n = O_P(\kappa_\Lambda\bar{\varepsilon}_n)$ suffices below.

Step 3: empirical converges to population. $\widehat{\tau}_K(V, D)$ is (up to $O(1/n)$) a U-statistic of order four in the i.i.d. sample, so $\widehat{\tau}_K(V, D) \to \tau_K(D, V)$ in probability (imported, §5); the analogous consistency of the sample Spearman coefficient for continuous population laws is classical (§5), and the continuum laws here are atomless (Lemma 4 for D; for V, the population functionals are those of the Green-limit theorem, which are well-defined grade correlations there).

Assemble. On the intersection of the calibration event, the Step-2 event, and the Step-3 convergence, $\widehat{\tau}_K(\widehat{V}, D) \ge \tau_K(D, V) - 2f_n - o_P(1) \ge \tau_\star - C\kappa_\Lambda\bar{\varepsilon}_n - o_P(1)$, and identically for $\widehat{\rho}_S$ with the constant from Step 1. □

Remark 10 (The growing-window condition, rederived). At a growing cutoff, the angular distances concentrate: $V = \sqrt{2} + O(A_\Lambda^{-1})$ with the pair-dependence of size $\asymp A_\Lambda^{-1}$ (Paper I). Concentration forces $\kappa_\Lambda \gtrsim A_\Lambda$, and the natural scale-aware form of (AC) is the matching upper bound $\kappa_\Lambda \le C\,A_\Lambda$. Since $\rho_\Lambda^- \asymp \sqrt{A_\Lambda}$, the error term in Theorem 8 scales as

$$\kappa_\Lambda\,\bar{\varepsilon}_n \;\asymp\; A_\Lambda \cdot \frac{E_n}{\sqrt{A_\Lambda}} \;=\; E_n\sqrt{A_\Lambda},$$

so the transfer condition is $E_n\sqrt{A_m} \to 0$ — *exactly* the sharpened growing-window condition derived in Paper I by the direct kernel comparison. That two independent routes (kernel rescaling; anti-concentration counting) produce the same condition is evidence that $E_n\sqrt{A_m} \to 0$ is the correct calibration threshold, not an artifact of either proof.

Remark 11 (Status of (AC)). (AC) at fixed Λ asserts a bounded density for the pushforward of the sampling measure under $y \mapsto \|N_\Lambda(x) - N_\Lambda(y)\|$, uniformly in x. By the coarea formula this reduces to an integrated gradient lower bound $\int_{\{V(x,\cdot)=t\}} |\nabla_y V(x,y)|^{-1}$ being bounded, i.e. to $\nabla_y V$ not degenerating on large portions of level sets — eigenfunction-level technology (nodal/critical set estimates) is the natural tool, and a proof is a self-contained open problem. Two mitigations: (i) (AC) is directly measurable on any substrate (estimate the density of sampled V-values; the pre-registered pipeline can emit this per run), turning the hypothesis into a per-instance certificate; (ii) Theorem 8 survives with (AC) replaced by any bound $\mathbb{P}(|V_P - V_Q| \le \epsilon) \le \omega(\epsilon)$ with $\omega(2\bar{\varepsilon}_n) \to 0$ — bounded density is the simplest sufficient form, not the needed one.

Remark 12 (Graph geodesics on the D-side). The theorem compares against manifold distances $D = d_M$. Replacing D by graph-hop geodesics adds a D-side perturbation, handled by the identical Step-1/Step-2 argument once a uniform graph-geodesic consistency bound is available (standard for ε-graphs above the connectivity scale); the required anti-concentration on the D-side is automatic, since the pair-distance law on a closed manifold has a bounded density.

7 The anti-concentration hypothesis (AC′)

Purpose. The rank-transfer theorem (§2, "Rank transfer without bi-Lipschitz hypotheses") assumes an anti-concentration hypothesis to control the fraction of near-tied comparisons. It was stated as *bounded density*:

$$\text{(AC)} \qquad \kappa_\Lambda := \sup_{x \in M} \big\| \text{density of } V(x,Y) \big\|_\infty < \infty, \qquad V(x,y) := \|N_\Lambda(x) - N_\Lambda(y)\|,$$

Y drawn from the sampling density. This Part does two things. First, it shows that (AC) as bounded density is *generically false* in dimension 2: a nondegenerate saddle of $V(x,\cdot)$ forces a logarithmic divergence of the density. Second, it proves the *modulus* form that the transfer theorem actually consumes—and it is the correct hypothesis, exactly as §2 anticipated ("bounded density is the simplest sufficient form, not the needed one"). For real-analytic metrics the modulus form holds unconditionally, by a uniform Łojasiewicz sublevel estimate, so (AC) is discharged on the standard examples (tori, spheres, projective and homogeneous spaces, and their real-analytic deformations). Notation follows §2; the cutoff Λ is fixed and multiplet-closed, and N_Λ is the normalized truncated commute-weighted eigenmap.

8 Bounded density is the wrong hypothesis

Fix x and consider $V(x,\cdot) : M \to [0,2]$, $V(x,y)^2 = 2 - 2C_\Lambda(x,y)$ with $C_\Lambda(x,y) = \langle N_\Lambda(x), N_\Lambda(y)\rangle$. At a fixed cutoff N_Λ is a smooth map, so $V(x,\cdot)$ is smooth, and its pushforward density is governed by the coarea formula

$$f_{V(x,\cdot)}(t) = \int_{\{V(x,\cdot)=t\}} \frac{\rho(y)}{|\nabla_y V(x,y)|}\, d\mathcal{H}^{d-1}(y), \tag{2}$$

ρ the sampling density. The integrand blows up where $\nabla_y V = 0$, i.e. at critical points of $V(x,\cdot)$, and whether (2) stays bounded is decided by the *type* of those critical points.

Proposition 13 (Saddles break bounded density in $d = 2$)**.** *Let $d = 2$ and suppose $V(x,\cdot)$ has a nondegenerate (Morse) saddle at some $y_0 \neq x$, with $V(x,y_0) = t_0$ and $\rho(y_0) > 0$. Then*

$f_{V(x,\cdot)}(t) \to \infty$ as $t \to t_0$; indeed $f_{V(x,\cdot)}(t) \geq c \log \frac{1}{|t-t_0|}$ near t_0. In particular $\kappa_\Lambda = \infty$, so (AC) as bounded density fails.

Proof. In Morse coordinates $V(x,\cdot) = t_0 + \frac{1}{2}(\xi_1^2 - \xi_2^2) + o(|\xi|^2)$ near y_0, so level sets near t_0 are hyperbolas $\xi_1^2 - \xi_2^2 = 2(t-t_0)$ and $|\nabla V| \asymp |\xi|$. Integrating $|\xi|^{-1}$ along a level hyperbola over a fixed coordinate neighborhood gives a contribution $\asymp \int \frac{d\ell}{|\xi|} \asymp \log \frac{1}{|t-t_0|}$ (the standard logarithmic saddle divergence of the density of states). With $\rho(y_0) > 0$ the weight is bounded below, so $f_{V(x,\cdot)}(t) \gtrsim \log \frac{1}{|t-t_0|}$. □

Saddles are forced on positive genus: on a closed surface with Euler characteristic $\chi \leq 0$ (genus ≥ 1), $\#\text{min} - \#\text{saddle} + \#\text{max} = \chi$ with $\#\text{min}, \#\text{max} \geq 1$ gives $\#\text{saddle} \geq 2 - \chi \geq 2$, so a Morse $V(x,\cdot)$ always has saddles and bounded density fails—in particular on the flat tori of the paper's torus experiments. (On S^2, $\chi = 2$ permits a saddle-free height-type V; and on the round sphere $V(x,\cdot) = h(d_M(x,\cdot))$ is non-Morse, with degenerate critical *circles* where bounded density can still fail through a distinct mechanism, but is not forced by Morse theory. So the failure is not universal—it is guaranteed exactly where Morse theory forces saddles, $\chi \leq 0$.) Where it occurs the divergence is only logarithmic, so the *measure* of near-tied pairs is still $O(\varepsilon \log \frac{1}{\varepsilon})$; the transfer theorem, which needs only that measure, is unharmed. This is precisely why the modulus form below is the right object.

9 The modulus form, and what the transfer theorem needs

Definition 14 ((AC′), modulus form)**.** *N_Λ satisfies (AC′) with modulus ω if $\omega : [0,\infty) \to [0,\infty)$ is nondecreasing with $\omega(0^+) = 0$ and*

$$\sup_{x\in M} \sup_{t\in\mathbb{R}} \mathbb{P}_Y\big(|V(x,Y) - t| \leq \varepsilon\big) \;\leq\; \omega(\varepsilon) \qquad \textit{for all } \varepsilon > 0.$$

Lemma 15 ((AC′) is what the transfer proof uses)**.** *In the proof of the rank-transfer theorem, every appearance of (AC) may be replaced by (AC′): the flip-fraction bound becomes $\mathbb{E}f_n \leq \omega(2\bar{\varepsilon}_n)$ (disjoint and anchor-sharing pairs-of-pairs alike), and the conclusion holds with the error term $C\kappa_\Lambda\bar{\varepsilon}_n$ replaced by $\omega(2\bar{\varepsilon}_n)$. Consequently the rank correlations satisfy $\widehat{\tau}_K, \widehat{\rho}_S \geq \tau_\star, \rho_\star - C\,\omega(2\bar{\varepsilon}_n) - o_P(1)$, and the transfer holds whenever $\omega(2\bar{\varepsilon}_n) \to 0$.*

Proof. The only role of (AC) in the transfer proof is Step 2, which bounds, for a pair-of-pairs $\{P,Q\}$, the probability $\mathbb{P}(|V_P - V_Q| \leq 2\bar{\varepsilon}_n)$. For vertex-disjoint $\{P,Q\}$, condition on $V_Q = t$ and apply the inner bound to $V_P = V(X,Y)$: averaging (2)'s tail, $\mathbb{P}(|V_P - t| \leq 2\bar{\varepsilon}_n) \leq \sup_{x,t} \mathbb{P}_Y(|V(x,Y) - t| \leq 2\bar{\varepsilon}_n) \leq \omega(2\bar{\varepsilon}_n)$ by (AC′) after further conditioning on the free endpoint of P. For anchor-sharing $P = (x_i,x_j), Q = (x_i,x_k)$, condition on x_i and V_Q: given x_i, $V_P = V(x_i,x_j)$ with x_j independent, and (AC′) at the anchor x_i gives the same bound—this is exactly the uniform-in-x (equivalently uniform-in-anchor) form. Hence $\mathbb{E}f_n \leq \omega(2\bar{\varepsilon}_n)$, and Markov + the Diaconis–Graham/Spearman conversion of Step 1 carry the rest verbatim. □

10 (AC′) holds for real-analytic metrics

Theorem 16 (Uniform Łojasiewicz modulus)**.** *Let M be a closed real-analytic Riemannian manifold and Λ a fixed multiplet-closed cutoff for which N_Λ separates points (so $V(x,\cdot)$ is non-constant for every x; this holds past the immersion threshold). Assume the sampling density ρ*

is bounded. Then there are constants $C_\Lambda < \infty$ and $\theta_\Lambda \in (0, 1]$, depending only on (M, Λ), with

$$\sup_{x\in M}\ \sup_{t\in\mathbb{R}}\ \mathrm{vol}\big\{y : |V(x,y) - t| \le \varepsilon\big\} \ \le\ C_\Lambda\, \varepsilon^{\theta_\Lambda} \qquad (\varepsilon > 0),$$

so (AC′) holds with $\omega(\varepsilon) = \|\rho\|_\infty C_\Lambda \varepsilon^{\theta_\Lambda}$. In particular $\omega(2\bar{\varepsilon}_n) = O(\bar{\varepsilon}_n^{\theta_\Lambda}) \to 0$, and by Lemma 15 the rank-transfer theorem holds with (AC) discharged.

Proof. On a real-analytic manifold with a real-analytic metric the Laplace eigenfunctions are real-analytic (elliptic regularity for operators with real-analytic coefficients), so the finite combinations $G_\Lambda(x, \cdot) = \sum_{0<\mu_k\le\Lambda} \mu_k^{-1}\varphi_k(x)\varphi_k(\cdot)$ and $S_\Lambda = G_\Lambda(\cdot, \cdot)$ are real-analytic, and hence so is

$$u_x(y) := V(x,y)^2 = 2 - \frac{2G_\Lambda(x,y)}{\sqrt{S_\Lambda(x)S_\Lambda(y)}},$$

jointly real-analytic in $(x, y) \in M \times M$ (the denominator is bounded below: by Hörmander's uniform local Weyl law the pointwise spectral function satisfies $e(\Lambda, x) > 1$ for all x once $\Lambda \ge \Lambda_1(M)$, so at every point some nonconstant mode $\le \Lambda$ is nonzero and $S_\Lambda(x) > 0$). For each fixed x, u_x is a non-constant real-analytic function on the compact M (non-constant because N_Λ separates points), so by the Łojasiewicz sublevel-set volume estimate there are $C(x), \theta(x)$ with $\mathrm{vol}\{|u_x - s| \le \delta\} \le C(x)\delta^{\theta(x)}$ uniformly in s.

For the uniformity over x we do not appeal to semicontinuity of the exponent (which is not, in general, enough to keep both $\theta(x)$ bounded below and $C(x)$ bounded above); instead we use o-minimality. Consider the joint concentration function

$$\mathcal{G}(x, \delta) := \sup_s \mathrm{vol}\big\{y : |u_x(y) - s| \le \delta\big\}.$$

Since $(x, y) \mapsto u_x(y)$ is real-analytic on the compact $M \times M$, the fiber volume of the subanalytic family $\{(x, y, s) : |u_x(y)-s| \le \delta\}$ is a log-analytic function of its parameters (Comte–Lion–Rolin), hence $\mathcal{G}$ is definable in the o-minimal structure $\mathbb{R}_{\mathrm{an,exp}}$; and $\mathcal{G}(x, 0) = 0$ for every x, because each non-constant real-analytic u_x has null level sets. The o-minimal Łojasiewicz inequality, applied on the compact base $M \times [0, \delta_0]$ to the definable $\mathcal{G} \ge 0$ vanishing on the definable set $\{\delta = 0\}$, yields a single $\theta_\Lambda \in (0, 1]$ and $C_\Lambda < \infty$ with $\mathcal{G}(x, \delta) \le C_\Lambda\, \delta^{\theta_\Lambda}$ uniformly in x; that is, $\sup_x \sup_s \mathrm{vol}\{|u_x - s| \le \delta\} \le C_\Lambda \delta^{\theta_\Lambda}$.

Transfer to $V = \sqrt{u}$. On $\{V \ge t_1 > 0\}$ the map $u \mapsto \sqrt{u}$ is bi-Lipschitz, so $\{|V - t| \le \varepsilon\} \subseteq \{|u - t^2| \le C'\varepsilon\}$ for $t \ge t_1$ and $\mathrm{vol} \le C_\Lambda(C'\varepsilon)^{\theta_\Lambda}$. Near $V = 0$ ($t < t_1$): $V = 0$ only where $N_\Lambda(y) = N_\Lambda(x)$; since N_Λ separates points this is the single point $y = x$ (or a subanalytic set of dimension $< d$, hence measure zero), and the sublevel $\{V \le t_1\}$ has $\mathrm{vol}\{|V - t| \le \varepsilon\} \le \mathrm{vol}\{u \le (t_1 + \varepsilon)^2\} \le C_\Lambda(t_1 + \varepsilon)^{2\theta_\Lambda}$; choosing $t_1 \asymp \varepsilon^{1/2}$ absorbs this into the same power law up to a constant. Multiplying by $\|\rho\|_\infty$ converts volume to probability, giving (AC′) with $\omega(\varepsilon) = \|\rho\|_\infty C_\Lambda \varepsilon^{\theta_\Lambda}$ (after adjusting C_Λ). The transfer-theorem consequence is Lemma 15. □

Remark 17 (The exponent, and the two mitigations retained)*.* θ_Λ is the reciprocal of the worst Łojasiewicz multiplicity of the family $\{u_x\}$; for Morse u_x one has $\theta_\Lambda = 1$ in $d \ge 3$ and $\theta_\Lambda = 1$ away from saddles with a logarithmic correction at saddle values in $d = 2$ (matching Proposition 13: the modulus is $\varepsilon \log(1/\varepsilon)$ there, still $\to 0$). Both mitigations noted in §2 survive and are strengthened: (i) ω is directly measurable per substrate (estimate the empirical modulus of sampled V-values), turning (AC′) into a per-instance certificate with a proven a-priori form on analytic geometries; (ii) only $\omega(2\bar{\varepsilon}_n) \to 0$ is needed, and $\omega(\varepsilon) = C\varepsilon^\theta$ delivers a definite convergence *rate* $O(\bar{\varepsilon}_n^{\theta_\Lambda})$, sharper than the qualitative statement.

Remark 18 (Growing cutoff)*.* Theorem 16 is at fixed Λ; the constant C_Λ degrades as $\Lambda \to \infty$ because the angular distances concentrate at $\sqrt{2}$. Quantitatively the density scale is $\kappa_\Lambda \asymp A_\Lambda$ (§2, Remark on the growing window), i.e. $\omega_\Lambda(\varepsilon) \asymp A_\Lambda \varepsilon$ in the regime where the Morse structure is nondegenerate, so $\omega_\Lambda(2\bar{\varepsilon}_n) \asymp A_\Lambda \bar{\varepsilon}_n \asymp E_n\sqrt{A_m}$ (using $\bar{\varepsilon}_n = 4E_n/\rho_\Lambda^-$ and $\rho_\Lambda^- \asymp \sqrt{A_\Lambda}$). This reproduces the sharpened growing-window threshold $E_n\sqrt{A_m} \to 0$ a third time—now from the analytic-modulus side—consistent with the kernel-rescaling and anti-concentration-counting derivations already on record.

11 The unconditional growing window and smooth-metric atomlessness

This part sharpens the transfer of Part 2 and the anti-concentration of Part 7 to three results. The transfer consumes only *pairwise* anti-concentration, so the uniform-in-anchor clause is deleted (§11.1); at a growing spectral cutoff the hypothesis is not merely weakened but *discharged*, turning the growing-window rank form into an *equivalence* with Green-rank positivity (§11.2); and at a fixed cutoff, atomlessness—hence (AC$'$)—holds for every *smooth* metric satisfying an explicit spectral nondegeneracy, automatic in the analytic category (§11.3). Two structural lemmas on $-G$ then point at the sole remaining link, Green-rank positivity (§11.4).

11.1 Anchor elimination: the transfer needs only the pair law

The rank-transfer theorem (Theorem 8) and its modulus form (Lemma 15) impose anti-concentration *uniformly in the anchor* x, because Step 2 of the proof treats anchor-sharing pairs-of-pairs by conditioning on the shared vertex. This clause can be deleted.

Theorem 19 (Anchor elimination)**.** *In the setting of Theorem 8, define the* pair modulus

$$\mu_\Lambda(\varepsilon) \;:=\; \mathbb{P}\big(|V_P - V_Q| \le \varepsilon\big), \qquad \textit{P, Q independent pairs from the sampling density.}$$

Then the conclusion of Theorem 8 holds with the error term $C\kappa_\Lambda\bar{\varepsilon}_n$ replaced by $C\,\mu_\Lambda(2\bar{\varepsilon}_n) + O(1/n)$: no anchored hypothesis of any kind is required. Consequently (AC$'$) *may be restated as: $\mu_\Lambda(\varepsilon) \le \omega(\varepsilon)$ with $\omega(0^+) = 0$, and the transfer holds whenever $\mu_\Lambda(2\bar{\varepsilon}_n) \to 0$.*

Proof. Among the $\binom{N_p}{2}$ pairs-of-pairs formed from the $N_p = \binom{n}{2}$ sampled pairs, those sharing a vertex number at most $3n^3$, a fraction $O(1/n)$ of the total $\asymp n^4$. Bound their flip indicator trivially by 1: they contribute $O(1/n)$ to the flip fraction f_n, hence $O(1/n)$ to the Kendall and Spearman perturbations of Step 1, which survive verbatim. For vertex-disjoint pairs-of-pairs, P and Q are independent draws of the pair, so $\mathbb{E}[\text{flip indicator}] \le \mathbb{P}(|V_P - V_Q| \le 2\bar{\varepsilon}_n) = \mu_\Lambda(2\bar{\varepsilon}_n)$ directly—no conditioning on a shared anchor ever occurs. Markov's inequality and Steps 1 and 3 of the original proof are unchanged. □

Remark 20. Two immediate consequences. (i) the modulus form (Definition 14, Lemma 15) simplifies: the $\sup_x$ is deleted, and the modulus is a property of the single *pair law* of $V_\Lambda(X,Y)$. (ii) Since the concentration function of the difference of two i.i.d. copies of any random variable tends, as $\varepsilon \to 0$, to the sum of squared atom masses, one has $\mu_\Lambda(\varepsilon) \to 0 \iff$ the law of $V_\Lambda(X,Y)$ is *atomless*. Anti-concentration, in the only form the transfer consumes, *is* atomlessness of the pair law. Everything below exploits this.

11.2 The growing window, unconditionally

The one analytic input is a uniform relative Weyl law for the commute radius; we state it as a lemma rather than invoke it in passing.

Lemma 21 (Uniform relative Weyl law for the commute radius)**.** *Let M be closed and smooth, $2 \le d \le 3$, and $S_\Lambda(x) = \sum_{0<\mu_k\le\Lambda} \mu_k^{-1}\varphi_k(x)^2$ the squared commute radius. Then $S_\Lambda(x) = A_\Lambda\big(1+r_\Lambda(x)\big)$ with $A_\Lambda = \sum_{0<\mu_k\le\Lambda}\mu_k^{-1}$ and $\rho_\Lambda := \sup_x|r_\Lambda(x)| \to 0$; quantitatively $\rho_\Lambda = O(1/\log\Lambda)$ for $d=2$ and $O(\log\Lambda/\sqrt{\Lambda})$ for $d=3$. Consequently the radius floor $\rho_\Lambda^- = \sqrt{\inf_x S_\Lambda} = \sqrt{A_\Lambda}\,(1+o(1))$.*

Proof. Let $e(\lambda,x) = \sum_{\mu_k\le\lambda}\varphi_k(x)^2$ be the pointwise spectral function. Hörmander's *uniform* pointwise Weyl law [13] gives $e(\lambda,x) = c_d\lambda^{d/2} + R(\lambda,x)$ with $\sup_x|R(\lambda,x)| \le C\lambda^{(d-1)/2}$. Excluding the constant mode ($\mu_1 > 0$), Stieltjes integration by parts yields

$$S_\Lambda(x) = \int_{\mu_1^-}^{\Lambda} \lambda^{-1}\,d_\lambda e(\lambda,x) = \frac{e(\Lambda,x)}{\Lambda} + \int_{\mu_1}^{\Lambda}\frac{e(\lambda,x)}{\lambda^2}\,d\lambda.$$

The main term $c_d\lambda^{d/2}$ contributes the x-independent piece of size $c_2\log\Lambda$ ($d=2$) or $c_3'\sqrt{\Lambda}$ ($d=3$), equal to $A_\Lambda(1+o(1))$ (the same integration by parts applied to $dN(\lambda)$). The remainder is bounded *uniformly in x* by $\Lambda^{-1}|R(\Lambda,x)| + \int_{\mu_1}^{\Lambda}\lambda^{-2}|R(\lambda,x)|\,d\lambda \le C\Lambda^{(d-3)/2} + C\int_{\mu_1}^{\Lambda}\lambda^{(d-5)/2}\,d\lambda$, which is $O(1)$ for $d=2$ and $O(\log\Lambda)$ for $d=3$. Dividing the uniform remainder by the diverging main term gives the stated ρ_Λ; the floor follows since $\inf_x S_\Lambda = A_\Lambda(1-\rho_\Lambda)$. The hypothesis $2\le d\le 3$ is used here: at $d=1$, $A_\Lambda = \sum\mu_k^{-1} < \infty$ converges and $S_\Lambda(x) \to G(x,x)$ is a non-constant limit, so no relative Weyl law holds—but d=1 is trivial for the theorem below ($\rho_G \equiv 1$). □

Theorem 22 (Growing-window rank transfer for all smooth metrics)**.** *Let M be a closed smooth Riemannian manifold, $2\le d\le 3$ (the case d=1 is trivial: $\rho_G\equiv 1$), volume normalized, with i.i.d. samples from a density bounded above and below. Let $\Lambda_n\to\infty$ be multiplet-closed cutoffs and suppose the calibration event of Theorem 8 holds with*

$$E_n\sqrt{A_{\Lambda_n}} \;\longrightarrow\; 0.$$

Then the empirical rank correlations of the graph angular distances against the manifold distances converge in probability to the Green-rank coefficient:

$$\widehat{\rho}_S\big(\widehat{V},D\big) \;\xrightarrow{\mathbb{P}}\; \rho_G(M), \qquad \widehat{\tau}_K\big(\widehat{V},D\big) \;\xrightarrow{\mathbb{P}}\; \tau_G(M).$$

No anti-concentration hypothesis is assumed: it is proved.

Proof. By Theorem 19 and Step 1 of the transfer proof it suffices to show $\mu_{\Lambda_n}(2\bar{\varepsilon}_n)\to 0$, where $\bar{\varepsilon}_n = 4E_n/\rho_{\Lambda_n}^-$; the empirical-to-population step is uniform over n because the grade kernels are bounded, so the U-statistic variance bounds are kernel-independent, and the population coefficients converge to ρ_G,τ_G by the Green-kernel rank-limit theorem.

Step 1: from V to the truncated Green kernel. Since $V_P - V_Q = (V_P^2 - V_Q^2)/(V_P+V_Q)$ and $0 < V_P + V_Q \le 4$,

$$\{|V_P - V_Q| \le 2\bar{\varepsilon}_n\} \;\subseteq\; \{|\widetilde{C}_P - \widetilde{C}_Q| \le 4\bar{\varepsilon}_n\}, \qquad \widetilde{C} := \tfrac{1}{2}(2-V^2) = K_\Lambda/\sqrt{S_\Lambda S_\Lambda}.$$

Lemma 21 gives $S_\Lambda(x) = A_\Lambda(1 + r_\Lambda(x))$ with $\rho_\Lambda := \sup_x |r_\Lambda(x)| \to 0$, hence pointwise

$$A_\Lambda \widetilde{C}(x,y) = K_\Lambda(x,y)\big(1 + O(\rho_\Lambda)\big), \qquad A_\Lambda \bar{\varepsilon}_n \le 5\, E_n \sqrt{A_{\Lambda_n}} =: \delta_n \to 0.$$

The multiplicative error is *relative* to K_Λ, which is unbounded near the diagonal, so truncate: for $R > 0$,

$$\mu_{\Lambda_n}(2\bar{\varepsilon}_n) \;\le\; \mathbb{P}\big(|K_P - K_Q| \le 2\delta_n + C\rho_{\Lambda_n} R\big) \;+\; 2\,\mathbb{P}\big(|K_\Lambda(X,Y)| > R\big),$$

and $\mathbb{P}(|K_\Lambda| > R) \le C/R$ uniformly in Λ, since $\mathbb{E}|K_\Lambda| \le \|K_\Lambda\|_{L^2} \le \|G\|_{L^2} + o(1)$ (bounded density; $d \le 3$).

Step 2: comparison with the Green pair law. With $T_\Lambda := \|K_\Lambda - G\|^2_{L^2(\mathrm{vol}^{\otimes 2})} = \sum_{\mu_k > \Lambda} \mu_k^{-2} \to 0$ ($d \le 3$), Chebyshev gives, for any $\gamma > 0$ and $\beta_n := 2\delta_n + C\rho_{\Lambda_n} R$,

$$\mathbb{P}\big(|K_P - K_Q| \le \beta_n\big) \;\le\; \mathbb{P}\big(|G_P - G_Q| \le \beta_n + 2\gamma\big) \;+\; \frac{2C\, T_{\Lambda_n}}{\gamma^2}.$$

Step 3: the Green pair law is atomless, so its difference anti-concentrates. By Lemma 4 the law of $G(X,Y)$ is atomless; hence $Q(\varepsilon) := \mathbb{P}(|G_P - G_Q| \le \varepsilon) \downarrow \mathbb{P}(G_P = G_Q) = \sum (\text{atoms})^2 = 0$ as $\varepsilon \downarrow 0$.

Assemble. Choose $R = R_n = \rho_{\Lambda_n}^{-1/2} \to \infty$ and $\gamma = \gamma_n = T_{\Lambda_n}^{1/3} \to 0$; then $\beta_n \to 0$ and

$$\mu_{\Lambda_n}(2\bar{\varepsilon}_n) \;\le\; Q\big(\beta_n + 2\gamma_n\big) + 2C\, T_{\Lambda_n}^{1/3} + \frac{2C}{R_n} \;\longrightarrow\; 0.$$

All constants depend only on (M, g) and the density bounds. $\square$

Corollary 23 (The conjecture is the geometry: an equivalence)**.** *Let M be closed and smooth, $2 \le d \le 3$, with sampling and calibration as above along an admissible growing schedule (d=1 is trivial, $\rho_G \equiv 1$). Then*

$$\text{Angular Preservation (growing-window rank form) holds on } M \quad \Longleftrightarrow \quad \rho_G(M) > 0,$$

and when it holds, the optimal floor is $\rho_0(M) = \rho_G(M)$. In particular the conjecture, over any class $\mathcal{C}$ of smooth closed manifolds of dimension $2 \le d \le 3$, is equivalent to the single potential-theoretic statement $\inf_{M \in \mathcal{C}}$*-free positivity: $\rho_G(M) > 0$ for every $M \in \mathcal{C}$.*

Proof. Theorem 22 shows the empirical coefficient converges to $\rho_G(M)$: if $\rho_G(M) > 0$ the rank form holds with floor $\rho_G(M) - o(1)$; if $\rho_G(M) \le 0$ no positive floor is possible. $\square$

Remark 24. This supersedes Remark 10 (whose growing-window discussion was explicitly heuristic) and closes frontier item on the growing window: the sharpened threshold $E_n \sqrt{A_m} \to 0$, derived twice before by scaling arguments, now carries a complete proof, and the analyticity hypothesis plays no role at growing cutoffs. The mechanism is worth naming: at scale A_Λ^{-1} the angular pair statistic *is* the truncated Green kernel, and the anti-concentration of the limit—automatic from atomlessness, which is itself a theorem (Lemma 4)—is inherited back along the L^2 convergence, with the free-parameter split absorbing all rates. No structure of V_Λ at fixed Λ is ever used.

11.3 Fixed cutoffs: smooth metrics and the nondegeneracy (ND)

At a fixed multiplet-closed Λ the limit-comparison of Theorem 22 is unavailable; by Theorem 19 what is needed is exactly atomlessness of the pair law of $\widetilde{C}_\Lambda(X,Y) = K_\Lambda/\sqrt{S_\Lambda S_\Lambda}$. Write $\mu_1 < \cdots < \mu_J$ for the distinct nonzero eigenvalues $\leq \Lambda$ and $\Phi_j(x,y) = \sum_{\mu_k=\mu_j} \varphi_k(x)\varphi_k(y)$ for the multiplet kernels, so $K_\Lambda = \sum_j \mu_j^{-1}\Phi_j$ and $\Delta_y \Phi_j = \mu_j \Phi_j$.

Definition 25 (Spectral nondegeneracy)**.** *(M, g, Λ) satisfies* (ND) *if for each $j \leq J$ the set*

$$\mathrm{Flat}_j \;:=\; \bigl\{y \in M : \;\Delta^m \sqrt{S_\Lambda}\,(y) = \mu_j^m \sqrt{S_\Lambda}\,(y) \quad \forall m \geq 0\bigr\}$$

has volume zero.

Theorem 26 (Atomlessness of the angular pair law)**.** *Let M be closed and smooth (any d), Λ multiplet-closed with $S_\Lambda > 0$ and N_Λ separating points, and let the sampled pair (X, Y) have a* joint *density bounded above on $M \times M$ (so any atom of the scalar law of $\widetilde{C}_\Lambda(X,Y)$ forces a positive-measure level set—this is the only role the density plays, and it is not assumed bounded below). Then:*

(a) the level $\{\widetilde{C}_\Lambda = 0\}$ is null unconditionally;

(b) under (ND)*, the law of $\widetilde{C}_\Lambda(X,Y)$ is atomless; hence by Theorem 19 the fixed-cutoff rank transfer holds with* (AC) *discharged;*

(c) (ND) *holds automatically when g is real-analytic (recovering the qualitative content of Theorem 16 without Łojasiewicz or o-minimality); more generally, any atom of the pair law forces* $\mathrm{vol}(\mathrm{Flat}_j) > 0$ *for some j: a fixed smooth function $(\Delta - \mu_j)\sqrt{S_\Lambda}$, tuned to an exact eigenvalue, vanishing to infinite Δ-order on a set of positive measure.*

Proof. Throughout we use the *iterated level-set lemma*: if h is smooth and $E \subseteq \{h = 0\}$, then a.e. on E every partial derivative of h of every order vanishes. (Induction on the order: for C^1 functions $\nabla h = 0$ a.e. on $\{h = 0\}$ (Evans–Gariepy); apply this successively to each $\partial^\alpha h$, which vanishes a.e. on E by the previous step; a countable union of null sets is null.) Consequently, at a.e. point of E, *every* differential operator applied to h vanishes.

Suppose the pair law has an atom at τ, i.e. $E := \{(x,y) : \widetilde{C}_\Lambda(x,y) = \tau\}$ has positive $\mathrm{vol}^{\otimes 2}$ measure, and set $w(x,y) := K_\Lambda(x,y) - \tau\sqrt{S_\Lambda(x)}\,\sqrt{S_\Lambda(y)}$, which vanishes on E. By the iterated lemma, at a.e. $(x,y) \in E$ all mixed partials of w vanish; evaluating $\Delta_x^N w$ there for $N \geq 1$ and using $\Delta_x^N K_\Lambda = \sum_j \mu_j^{N-1}\Phi_j$:

$$\sum_{j=1}^{J} \mu_j^{N-1}\,\Phi_j(x,y) \;=\; \tau\,\bigl(\Delta^N \sqrt{S_\Lambda}\bigr)(x)\,\sqrt{S_\Lambda}\,(y), \qquad N = 1,\ldots,J, \text{ a.e. } (x,y) \in E. \tag{3}$$

The coefficient matrix $(\mu_j^{N-1})_{N,j}$ is a Vandermonde matrix with distinct nodes, hence invertible; solving,

$$\Phi_j(x,y) \;=\; \tau\,G_j(x)\,\sqrt{S_\Lambda}\,(y) \quad \text{a.e. on } E, \qquad G_j := \textstyle\sum_{N=1}^{J} a_{jN}\,\Delta^N\sqrt{S_\Lambda}, \tag{4}$$

with fixed smooth G_j determined by $\{\mu_i\}$ alone.

Case $\tau = 0$ (part (a)). Then $\Phi_j = 0$ a.e. on E for every j. By Fubini there is a positive-measure set of x for which $\Phi_j(x,\cdot)$ vanishes on a set of positive y-measure. But $\Phi_j(x,\cdot)$ lies in the

μ_j-eigenspace; a nonzero eigenfunction cannot vanish on a set of positive measure (by the iterated lemma it would vanish to infinite order at a point, contradicting Aronszajn's strong unique continuation (smooth-coefficient case; the Lipschitz-metric extension holds via Garofalo–Lin but is not needed here). Hence $\Phi_j(x,\cdot) \equiv 0$ for such x, i.e. $\varphi_k(x) = 0$ for every k in the multiplet (linear independence), for all j; so S_Λ vanishes on a positive-measure set of x, contradicting $S_\Lambda > 0$. (This uses that $S_\Lambda = \sum_{0<\mu_k\le\Lambda} \mu_k^{-1}\varphi_k^2$ *omits* the constant mode: the multiplets $j = 1,\dots,J$ exhaust the nonzero modes $\le \Lambda$, so their simultaneous vanishing genuinely forces $S_\Lambda = 0$—were $\varphi_0^2 = 1/\operatorname{vol} M$ retained, this step would fail.)

Case $\tau \neq 0$ (part (b)). Define $h_j := \Phi_j - \tau\, G_j \otimes \sqrt{S_\Lambda}$ on $M \times M$; by (4), $h_j = 0$ a.e. on E, so by the iterated lemma all its derivatives vanish a.e. on E; evaluating $\Delta_y^m h_j$ and using $\Delta_y^m \Phi_j = \mu_j^m \Phi_j$ together with (4):

$$\tau\, G_j(x) \left[\mu_j^m \sqrt{S_\Lambda}\,(y) - \Delta^m \sqrt{S_\Lambda}\,(y)\right] = 0 \qquad \text{a.e. } (x,y) \in E, \quad \forall m \ge 0,\ \forall j.$$

Fix j. Either the second factor vanishes for all m on a positive-measure set of y appearing in E—i.e. $\operatorname{vol}(\mathrm{Flat}_j) > 0$, excluded by (ND)—or $G_j(x) = 0$ for a.e. $(x,y) \in E$ (the exceptional set being null in the product measure on E). Under (ND) the second alternative holds for every j, and then (4) gives $\Phi_j = 0$ a.e. on E for every j: the argument of case $\tau = 0$ applies verbatim and yields a contradiction. Hence no atom exists.

Part (c). If g is real-analytic, so are the eigenfunctions, S_Λ, and $\sqrt{S_\Lambda}$ ($S_\Lambda > 0$); hence $\xi_j := (\Delta - \mu_j)\sqrt{S_\Lambda}$ is real-analytic. Now $\mathrm{Flat}_j \subseteq \{\xi_j = 0\}$ (its m=1 clause), so a positive-measure Flat_j would put the zero set of the analytic ξ_j at positive measure; on a connected analytic manifold the zero set of a non-identically-zero analytic function is null, forcing $\xi_j \equiv 0$. But $\xi_j \equiv 0$ makes $\sqrt{S_\Lambda}$ a *positive* eigenfunction with eigenvalue $\mu_j > 0$, impossible on a closed manifold (a positive eigenfunction forces the bottom of the spectrum, $\mu = 0$). Hence $\operatorname{vol}(\mathrm{Flat}_j) = 0$ for every j: (ND) holds. The final sentence of (c) is the case analysis just performed, read contrapositively. □

Remark 27 (What (ND) is, and is not). (ND) is a per-(M, g, Λ) condition on one fixed smooth function per distinct eigenvalue, violated only if $(\Delta - \mu_j)\sqrt{S_\Lambda}$ is infinitely Δ-flat on a fat set at the *exact* spectral value μ_j—a closed, nowhere-dense degeneracy in any reasonable topology on metrics, automatic in the analytic category, and checkable in principle. We do not currently know whether it can fail for some smooth metric; if it can, the corresponding atom is confined to the specific levels τ produced by the case analysis. Removing (ND) outright is the residue of frontier item (a); the item is otherwise closed.

11.4 Structure of the Green kernel: two lemmas toward positivity

The remaining summit is Green-rank positivity (Corollary 23 makes it *equivalent* to the growing-window conjecture). Two structural facts.

Lemma 28 (Sublevels of $-G$ cluster at the pole)**.** *Let M be closed and smooth, $x \in M$. With the geometer's sign convention $\Delta_{\mathrm{LB}} G(x,\cdot) = +1$ on $M \setminus \{x\}$, the function $-G(x,\cdot)$ is strictly superharmonic off the pole; consequently, for every $c \in \mathbb{R}$, each connected component of $\{y : -G(x,y) < c\}$ has x in its closure, and $\{-G(x,\cdot) < c\} \cup \{x\}$ is connected.*

Proof. If a component U had $x \notin \overline{U}$, then $-G(x,\cdot)$ would be continuous on the compact $\overline{U}$, equal to c on ∂U and $< c$ inside, hence would attain an interior minimum; at an interior minimum

$\Delta_{\mathrm{LB}}(-G) \geq 0$, contradicting $\Delta_{\mathrm{LB}}(-G) = -1 < 0$ on $M \setminus \{x\}$. Adjoining x, which lies in every component's closure, connects the sublevel set. □

Lemma 29 (Two-dimensional conformal reduction; log-additivity). *Let (Σ, g) be a closed surface, $g = e^{2\varphi} g_0$ with g_0 of constant curvature,* $\mathrm{vol}_g = 1$*. Then G_g is the zero-mean Green kernel of the* weighted *manifold $(\Sigma, g_0,\ e^{2\varphi}\,\mathrm{vol}_0)$: the operator is Δ_{g_0}, only the measure changes. Moreover d_g and d_{g_0} satisfy $e^{\min\varphi}\, d_{g_0} \leq d_g \leq e^{\max\varphi}\, d_{g_0}$ (lower bound because $\|\cdot\|_g \geq e^{\min\varphi}\|\cdot\|_{g_0}$ along every path, upper bound by testing the g_0-geodesic), so the ratio d_g/d_{g_0} varies within a factor $e^{\mathrm{osc}\,\varphi}$; since the two-dimensional profile is logarithmic, this distortion enters the profile* additively*: for the uniformized profile $\Psi_0 = -\gamma_2 -$ (regular part of g_0),*

$$\left\| \Psi_0(d_{g_0}) - \Psi_0(d_g) \right\|_\infty \;\leq\; \frac{\mathrm{osc}\,\varphi}{2\pi} \;+\; C(g_0, \mathrm{osc}\,\varphi) \quad \text{on } \{d_{g_0} \leq r_0\},$$

finite up to the diagonal. Consequently, on every closed surface the master remainder (Theorem 32) is finite for a profile built from the uniformization: Δ_{rad}-type control on surfaces is a statement about the weighted flat/round Green kernel and the C^0 oscillation of the conformal factor alone—no curvature bounds enter.

Proof. In dimension two $\Delta_g = e^{-2\varphi}\Delta_{g_0}$, so $\Delta_{g_0} G_g(x,\cdot) = e^{2\varphi}\big(\delta_x^{(g)} - 1\big)$-type identities reorganize into the weighted-Laplacian Green equation on $(\Sigma, g_0, e^{2\varphi}\,\mathrm{vol}_0)$: the conformal factor moves entirely into the measure. The displayed bound is the logarithmic profile's response to multiplicative distortion: $\log d_g - \log d_{g_0} \in [\min\varphi, \max\varphi]$ has oscillation $\leq \mathrm{osc}\,\varphi$, and the zero-mean profile absorbs the (immaterial) additive constant, leaving $\mathrm{osc}\,\varphi/2\pi$ plus the (bounded) distortion of the regular part of Ψ_0 over the compact off-diagonal range. □

Remark 30 (The remaining crux, stated exactly). After Theorems 22 and 26, the growing-window conjecture *is* the statement $\rho_G(M) > 0$ (Corollary 23). What Lemmas 28–29 contribute: the sublevels of $-G(x,\cdot)$ are topologically ball-like around the pole (no isolated far minima), so any failure of positivity must come from *elongated* sublevel geometry at unit scale; and on surfaces the entire problem conformally reduces to weighted constant-curvature Green kernels, where the diagonal is handled by log-additivity and the fight is confined to the top distance octave. The sharpest open formulation: for every closed surface, do the unit-scale sublevel sets of the weighted flat/round Green kernel order the bulk pairs with positive rank correlation? The numerical plateau ($\rho_G \gtrsim 0.79$ over four families) is evidence the answer is yes, possibly with a universal surface constant; a proof is the summit.

12 A non-radiality calculus for Green-rank positivity

Purpose. Paper I reduces the rank form of Angular Preservation, for $d \leq 3$, to *Green-rank positivity* (Conjecture, "Green-rank positivity"): $\rho_G(M) := \rho_S\big(d_M(X,Y), -G_M(X,Y)\big) > 0$, where G_M is the zero-mean Green kernel of the Laplace–Beltrami operator and (X,Y) are i.i.d. from the normalized volume. Two facts are known: $\rho_G = 1$ on compact two-point homogeneous spaces (Paper I, "Uniform rank preservation"), because there G is a strictly decreasing function of distance; and $\rho_G > 0$ on a C^2-open neighborhood of any metric with $\rho_G > 0$ (§2, Theorem 6). Both are qualitative and neither localizes the obstruction. This Part gives (i) a *master inequality* bounding the rank deficit $1-\rho_G$ by a single geometric quantity—the non-radiality of G, measured in sup norm against the best monotone radial profile—and (ii) an *explicit near-radial class* on which $\rho_G > 0$ with a quantitative bound. The two-point homogeneous result is the exact-radial ($R \equiv 0$) point of the master inequality, and the openness theorem is its small-R corollary made quantitative. Notation and imported facts follow Paper I and §2 (in particular Proposition "Distortion controls rank" and Lemma "Automatic atomlessness"); §26 states the standing assumptions and what remains open.

13 The master rank-remainder inequality

Fix M closed of dimension d, $\mathrm{vol}(M) = 1$, with zero-mean Green kernel G, and let (X, Y) be i.i.d. from the volume. Write $W = d_M(X, Y)$ for the random pairwise distance, $D = \operatorname{diam} M$, and let W' be an independent copy of W. All rank correlations are the population grade correlations. By Lemma 4 the laws of $d_M(X, Y)$ and of $G(X, Y)$ are atomless (the pair law has bounded density w.r.t. $\mathrm{vol} \otimes \mathrm{vol}$), so no ties occur and Daniels' inequality $|3\tau - 2\rho_S| \leq 1$ applies to every pair of the statistics below.

Definition 31. *A* monotone radial profile *is a strictly increasing continuous function $\Psi : (0, D] \to \mathbb{R}$. Its* remainder *is $R(x, y) := -G(x, y) - \Psi(d_M(x, y))$, and its* non-radiality *is $\|R\|_\infty = \sup_{x \neq y} |R(x, y)|$. Its* anti-concentration constant *is the essential sup of the density of the random variable $\Psi(W)$, denoted $\phi_\Psi := \|f_{\Psi(W)}\|_\infty \in (0, \infty]$.*

Theorem 32 (Master inequality)**.** *For every monotone radial profile Ψ with $\|R\|_\infty < \infty$ and $\phi_\Psi < \infty$,*

$$\rho_G(M) \;\geq\; 1 - 3\,\mu_\Psi\big(2\|R\|_\infty\big) \;\geq\; 1 - 12\,\phi_\Psi\,\|R\|_\infty, \qquad \mu_\Psi(\varepsilon) := \mathbb{P}\big(|\Psi(W) - \Psi(W')| \leq \varepsilon\big).$$

In particular $\rho_G(M) > 0$ whenever $\|R\|_\infty < 1/(12\,\phi_\Psi)$; and if $R \equiv 0$ (the kernel G is exactly a strictly decreasing function of distance), then $\rho_G(M) = 1$.

Proof. Let $P = (X, Y)$ and $Q = (X', Y')$ be two independent draws of the pair, with distances d_P, d_Q and Green values G_P, G_Q. Kendall's population coefficient for the pair of statistics $(d_M, -G)$ is $\tau_G = \mathbb{E}\big[\operatorname{sgn}(d_P - d_Q)\operatorname{sgn}(G_Q - G_P)\big] = 1 - 2p$, where $p = \mathbb{P}(\text{discordant})$ and a pair-of-pairs is discordant when $\operatorname{sgn}(d_P - d_Q) \neq \operatorname{sgn}\big((-G_P) - (-G_Q)\big)$; atomlessness rules out ties. By Daniels' inequality $\rho_S \geq (3\tau_G - 1)/2 = 1 - 3p$, i.e. $\rho_G \geq 1 - 3p$.

Bound p. Suppose $d_P < d_Q$ (the reverse case is symmetric). Discordance means $-G_P > -G_Q$, i.e. $\Psi(d_P) + R_P > \Psi(d_Q) + R_Q$, whence $\Psi(d_Q) - \Psi(d_P) < R_P - R_Q \leq 2\|R\|_\infty$. Since Ψ is strictly increasing and $d_P < d_Q$, the left side is positive, so $|\Psi(d_P) - \Psi(d_Q)| \leq 2\|R\|_\infty$. As $\Psi(d_P) = \Psi(W)$ and $\Psi(d_Q) = \Psi(W')$ are independent copies, $p \leq \mathbb{P}\big(|\Psi(W) - \Psi(W')| \leq 2\|R\|_\infty\big) = \mu_\Psi(2\|R\|_\infty)$. This gives the first inequality.

For the second, let $A = \Psi(W)$ with density $f_A \leq \phi_\Psi$. Then for any $\varepsilon \geq 0$, $\mu_\Psi(\varepsilon) = \mathbb{E}_{A'}\,\mathbb{P}\big(A \in [A' - \varepsilon, A' + \varepsilon]\big) \leq \mathbb{E}_{A'}\big[2\varepsilon\,\phi_\Psi\big] = 2\phi_\Psi\varepsilon$, so $\mu_\Psi(2\|R\|_\infty) \leq 4\phi_\Psi\|R\|_\infty$ and $\rho_G \geq 1 - 12\phi_\Psi\|R\|_\infty$. Finally if $R \equiv 0$ then $p = 0$ (the reverse ranking is exact), so $\tau_G = 1$ and $\rho_G = 1$. □

Remark 33. The two known results are the two extremes. *Two-point homogeneous*: $G(x, y) = g(d_M(x, y))$ with g strictly decreasing (Paper I), so $\Psi = -g$ has $R \equiv 0$ and Theorem 32 gives $\rho_G = 1$—recovering the corollary with no separate argument. *Openness*: $\|R\|_\infty$ and ϕ_Ψ vary continuously under C^2 perturbation of the metric (the Green kernel and the pair-distance law do), so a metric with $\rho_G = 1$ (hence $R \equiv 0$ for its optimal Ψ) has a C^2-neighborhood on which $\|R\|_\infty$ stays below $1/(12\phi_\Psi)$; Theorem 32 then gives $\rho_G > 0$ *with an explicit floor*, quantifying the qualitative openness theorem. What is genuinely new is that the inequality holds on *all* metrics, not only near the radial ones: it converts Green-rank positivity into a single sup-norm estimate on the non-radial part of G.

14 The near-diagonal singularity is radial to leading order

To turn Theorem 32 into a checkable geometric statement one needs a monotone radial profile Ψ whose remainder R is finite despite the diagonal blow-up of G. The parametrix supplies one: near the diagonal G equals the Euclidean fundamental profile plus a *bounded* regular part, and the singular coefficient is universal (independent of the point), so the diagonal contributes nothing to $\|R\|_\infty$.

Let γ_d be the Euclidean fundamental profile,

$$\gamma_3(r) = \frac{1}{4\pi r}, \qquad \gamma_2(r) = -\frac{1}{2\pi}\log r,$$

(and $\gamma_d(r) = \big((d-2)\omega_{d-1}\big)^{-1} r^{2-d}$ for $d \geq 3$ in general).

Lemma 34 (Bounded regular part)**.** *Let M be a closed smooth Riemannian d-manifold, $d \in \{2,3\}$, $\mathrm{vol}(M) = 1$, with injectivity radius $i_0 > 0$. Fix $r_0 < i_0$. There is a constant $H_0 = H_0(M, r_0) < \infty$ such that the zero-mean Green kernel satisfies*

$$\big|\, G(x,y) - \gamma_d\big(d_M(x,y)\big) \,\big| \;\leq\; H_0 \qquad \textit{whenever } d_M(x,y) \leq r_0.$$

Consequently, defining $\Psi(r) = -\gamma_d(r)$ for $r \leq r_0$ and extending Ψ to $[r_0, D]$ as any strictly increasing C^1 function with $\Psi' \geq c_1 > 0$ there and $\Psi(r_0^+) = \Psi(r_0^-)$, one has, with $H_{\mathrm{far}} := \sup_{d_M(x,y)\geq r_0}|G(x,y)| + \sup_{r\in[r_0,D]}|\Psi(r)|$,

$$\|R\|_\infty \;\leq\; \max\big(H_0,\; H_{\mathrm{far}}\big) \;<\; \infty,$$

and $\phi_\Psi < \infty$.

Proof. Regular part. In geodesic normal coordinates centered at y the metric is Euclidean to second order ($g^{ij} - \delta^{ij} = O(r^2)$, drift coefficients $O(r)$). Let χ be a cutoff, $\chi \equiv 1$ on $\{d_M(\cdot,y) \leq r_0\}$, supported in the i_0-ball. The parametrix $E_y(x) = \chi(x)\,\gamma_d(d_M(x,y))$ satisfies $\Delta_x E_y = \delta_y + b_y$, where $b_y = \big(\Delta_g - \Delta_{\mathrm{Eucl}}\big)\gamma_d \cdot \chi + [\Delta, \chi]\gamma_d$ collects the Euclidean-to-Riemannian discrepancy of the leading term (the $r^{2-d}/\log r$ singularity is annihilated by the Euclidean Laplacian) and the commutator with the cutoff (smooth, supported off the diagonal). Since $(\Delta_g - \Delta_{\mathrm{Eucl}})\gamma_d = O(r^{2-d})$, one has $b_y \in L^p(M)$ for every $p < \frac{d}{d-2}$ (all $p < \infty$ when $d = 2$), uniformly in y; this is where $d \leq 3$ enters. Then $h_y := G(\cdot,y) - E_y$ solves $\Delta h_y = -1 - b_y$ in the zero-mean class (from $\Delta_x G(\cdot,y) = \delta_y - 1$). Choose $p \in \big(\frac{d}{2}, \frac{d}{d-2}\big)$ —a nonempty range *exactly* for $d \in \{2,3\}$. Elliptic $W^{2,p}$ regularity and the Sobolev embedding $W^{2,p} \hookrightarrow C^0$ on the closed manifold give $\|h_y - \bar h_y\|_{C^0} \leq C\|{-1} - b_y\|_{L^p}$, where $\bar h_y = \int_M h_y = -\int_M \chi\,\gamma_d(d_M(\cdot,y))$ is bounded uniformly in y; hence $\|h_y\|_{C^0} \leq H_0$ uniformly. On $\{d_M(x,y) \leq r_0\}$, $\chi \equiv 1$ gives $E_y(x) = \gamma_d(d_M(x,y))$, so $|G(x,y) - \gamma_d(d_M(x,y))| = |h_y(x)| \leq H_0$.

Remainder bound. For $d_M(x,y) \leq r_0$, $R = -G + \gamma_d \circ d_M = -h_y$, bounded by H_0. For $d_M(x,y) \geq r_0$, both G (smooth off-diagonal, hence bounded on the compact set $\{d_M \geq r_0\}$) and $\Psi \circ d_M$ are bounded by H_{far}; hence $|R| \leq H_{\mathrm{far}}$ there.

Finiteness of ϕ_Ψ. Ψ is a strictly increasing C^1 diffeomorphism of $(0, D]$ onto its image, so $\Psi(W)$ has density $f_W(\Psi^{-1}(s))/\Psi'(\Psi^{-1}(s))$, where f_W is the density of $W = d_M(X, Y)$ (bounded, and $f_W(r) \asymp r^{d-1}$ as $r \to 0$). Near $s \to \Psi(0^+)$ $(r \to 0)$, $\Psi' = -\gamma_d' \asymp r^{1-d}$ $(d \geq 3)$ or r^{-1} $(d = 2)$, so the density $\asymp r^{d-1}/r^{1-d} = r^{2(d-1)} \to 0$ $(d \geq 3)$, resp. $r^{d-1}/r^{-1} = r^d \to 0$ $(d = 2)$; on $[r_0, D]$, $\Psi' \geq c_1 > 0$ and f_W is bounded, so the density is bounded. Hence $\phi_\Psi < \infty$. □

Remark 35. The universal leading coefficient is what makes the diagonal harmless: two pairs at nearly equal small distances have Green values agreeing to leading order regardless of *where* they sit, so the singular part contributes concordantly. All the non-radiality that can hurt the ranking lives in the bounded regular part h_y (the Robin function off its radial mean) and in the mesoscopic/large-scale values H_{far}—exactly the scales at which low eigenfunctions oscillate.

15 The explicit near-radial theorem

Theorem 36 (Green-rank positivity on a near-radial class)**.** *Let M be a closed smooth Riemannian d-manifold, $d \in \{2, 3\}$, $\mathrm{vol}(M) = 1$, injectivity radius $\geq i_0$. Let Ψ_0 be the parametrix profile of Lemma 34, with finite non-radiality $\|R_0\|_\infty$ and finite anti-concentration constant ϕ_0. Then*

$$\rho_G(M) \;\geq\; 1 - 12\,\phi_0\,\|R_0\|_\infty.$$

In particular $\rho_G(M) > 0$—so, by the Green-kernel rank-limit theorem (Paper I), $\liminf_\Lambda \rho_S\big(d_M(X, Y), \|N_\Lambda(X) - N_\Lambda(Y)\|\big) = \rho_G(M) \geq 1 - 12\phi_0\|R_0\|_\infty$*, the truncated commute angular ranking being uniform-in-mode-count rank-faithful with this floor—whenever*

$$\phi_0\,\|R_0\|_\infty \;<\; \frac{1}{12}.$$

Writing $\Delta_{\text{rad}}(M) := \inf_\Psi \|-G - \Psi \circ d_M\|_\infty$ for the best monotone-radial sup-approximation of $-G$ (the non-radiality of the Green kernel), one has $\|R_0\|_\infty \geq \Delta_{\text{rad}}(M)$, so a small non-radiality with controlled ϕ_0 suffices.

Proof. Immediate from Theorem 32 applied to the *single* profile Ψ_0, whose non-radiality $\|R_0\|_\infty$ and anti-concentration constant ϕ_0 are both finite by Lemma 34; no optimization over profiles is performed (in particular the bound does *not* split the product-infimum $\inf_\Psi \phi_\Psi \|R_\Psi\|_\infty$ into separate infima—an isotonic optimal profile can be flat, giving an atom in $\Psi(W)$ and $\phi_\Psi = \infty$). The rank-faithful consequence is the Green-kernel rank-limit theorem, which gives $\rho_S \to \rho_G$; the floor is thus approached as a lim inf up to $o(1)$, not attained at finite Λ. □

Remark 37 (What the class contains). On the two-point homogeneous spaces $-G$ is exactly monotone-radial ($R_0 \equiv 0$, hence $\Delta_{\text{rad}} = 0$ and $\rho_G = 1$); $\phi_0\|R_0\|_\infty$ is C^2-continuous, so the theorem covers a C^2-neighborhood of each with a numerical floor—the effective form of the openness theorem. More broadly, any manifold with $\phi_0\|R_0\|_\infty < 1/12$ is included: mild ellipsoids, low-eccentricity tori, small metric perturbations of symmetric spaces. Both $\|R_0\|_\infty$ and ϕ_0 are computable per metric (evaluate $-G - \Psi_0 \circ d_M$ and the density of $\Psi_0(W)$ over a volume sample), so the hypothesis is a certificate, not an article of faith. (The infimal non-radiality $\Delta_{\text{rad}} \leq \|R_0\|_\infty$ is a genuine geometric quantity, but it is *not* by itself sufficient: the sup-approximation optimal profile may be flat and carry $\phi = \infty$, so positivity is asserted through the fixed profile Ψ_0, whose ϕ_0 is finite.)

Remark 38 (Sharper constants via anti-concentration of $\Psi_0(W)$). The factor $12\phi_0$ is the crude one from bounding μ_{Ψ_0} by a uniform density. When the distance law is spread—$\Psi_0(W)$ far from concentrated—$\mu_{\Psi_0}(2\|R_0\|_\infty)$ is much smaller than $4\phi_0\|R_0\|_\infty$, and the first inequality of Theorem 32 ($\rho_G \geq 1 - 3\mu_{\Psi_0}$) is the honest bound to evaluate numerically. Manifolds with a well-spread pairwise-distance spectrum tolerate proportionally larger non-radiality.

16 A distributional refinement: the L^2 non-radiality bound

The master inequality controls ρ_G through $\|R\|_\infty$, the *sup* non-radiality. This is loose when $-G$ deviates from radial on only a small fraction of pairs: a short thin handle joining two distant regions makes $\|R\|_\infty$ large (the cross-handle pairs reverse) yet moves ρ_G negligibly, because those pairs are rare. The refinement below replaces the sup by the L^2 non-radiality, which is distribution-aware and, as it happens, scale-invariant—matching the fact that ρ_G is a rank correlation.

Definition 39. *For a monotone radial profile Ψ with remainder $R = -G - \Psi \circ d_M$, the L^2* non-radiality *is $V_\Psi := \mathbb{E}[R(X,Y)^2]$ and the* non-radiality index *is $\iota_\Psi := \phi_\Psi^2 V_\Psi$, with ϕ_Ψ the density bound of $\Psi(W)$, $W = d_M(X,Y)$. The index is dimensionless: ϕ_Ψ has units $[\Psi]^{-1}$ and V_Ψ units $[\Psi]^2$, so ι_Ψ is invariant under scaling the metric.*

Theorem 40 (Distributional master inequality)**.** *For every monotone radial profile Ψ with $\phi_\Psi < \infty$ and $V_\Psi < \infty$,*

$$\rho_G(M) \;\geq\; 1 - 18\left(\phi_\Psi^2 V_\Psi\right)^{1/3} \;=\; 1 - 18\,\iota_\Psi^{1/3}.$$

Hence $\rho_G(M) > 0$ whenever $\iota_\Psi < 18^{-3}$; and $V_\Psi = 0$ (radial $-G$) gives $\rho_G = 1$.

Proof. As in Theorem 32, $\rho_G \geq 1 - 3p$ with p the discordance probability, and a discordant pair-of-pairs with $d_P < d_Q$ forces $\Psi(d_Q) - \Psi(d_P) < R_P - R_Q \leq |R_P| + |R_Q|$; by symmetry $p \leq \mathbb{P}\big(|\Psi(d_P) - \Psi(d_Q)| < |R_P| + |R_Q|\big)$. Fix $\tau > 0$. Since $\{|\Psi(d_P) - \Psi(d_Q)| < |R_P| + |R_Q|\} \subseteq \{|\Psi(d_P) - \Psi(d_Q)| < \tau\} \cup \{|R_P| + |R_Q| \geq \tau\}$,

$$p \;\leq\; \mu_\Psi(\tau) + \mathbb{P}\big(|R_P| + |R_Q| \geq \tau\big) \;\leq\; 2\phi_\Psi\tau + 2\,\mathbb{P}\big(|R| \geq \tfrac{\tau}{2}\big),$$

using $\mu_\Psi(\tau) \leq 2\phi_\Psi\tau$ (Theorem 32) and a union bound with R_P, R_Q i.i.d. Chebyshev gives $\mathbb{P}(|R| \geq \tau/2) \leq 4V_\Psi/\tau^2$, so $p \leq 2\phi_\Psi\tau + 8V_\Psi/\tau^2$ and $\rho_G \geq 1 - 6\phi_\Psi\tau - 24V_\Psi/\tau^2$. Minimizing over τ at $\tau_\star = 2(V_\Psi/\phi_\Psi)^{1/3}$ gives $6\phi_\Psi\tau_\star + 24V_\Psi/\tau_\star^2 = 18(\phi_\Psi^2 V_\Psi)^{1/3}$. □

Remark 41 (Why it beats the sup bound). Always $V_\Psi \leq \|R\|_\infty^2$; the two bounds are complementary—take whichever is larger. Because of the cube-root exponent, the sup-norm master bound $1 - 12\phi_\Psi\|R\|_\infty$ is the better one in the *uniformly small* regime ($\phi_\Psi\|R\|_\infty \ll 1$), while Theorem 40 is *strictly* better exactly when the non-radiality is *concentrated*, $V_\Psi \ll \|R\|_\infty^2$. If $-G$ is non-radial by an amount $\asymp 1$ on a fraction ε of pairs and ≈ 0 elsewhere, then $\|R\|_\infty \asymp 1$ (the sup bound is vacuous) while $V_\Psi \asymp \varepsilon$, so $\iota_\Psi^{1/3} \asymp \varepsilon^{1/3} \to 0$: the L^2 bound stays sharp. This is exactly the thin-handle regime of the numerical hunt (`experiments/green-rank`): the handle spikes $\|R\|_\infty$ but leaves V_Ψ tiny, and ρ_G stays near 1—whereas a macroscopic bottleneck (a dumbbell neck) raises V_Ψ over an $\Omega(1)$ fraction of pairs and genuinely lowers ρ_G. The L^2 index is the quantity the experiments track.

Remark 42 (The constant, and the honest numerical bound)*.* The factor 18 comes from Chebyshev and is not sharp; controlling $\mathbb{P}(|R| \geq t)$ by a higher moment or by the actual tail of R improves it, and the pre-optimization form $\rho_G \geq 1 - 6\phi_\Psi\tau - 24V_\Psi/\tau^2$ (or the master $\rho_G \geq 1 - 3\mu_\Psi(2\|R\|_\infty)$) is the one to evaluate on a sample. The value of Theorem 40 is the *quantity*: a scale-invariant, distribution-aware non-radiality index that a sup-norm cannot express.

Remark 43 (Toward a geometric class)*.* Both factors are expected to admit geometric bounds: $\phi_\Psi \leq \Phi(K, D, i_0)$ from the bounded density of the pairwise-distance law under $\mathrm{Ric} \geq -(d-1)K$, $\mathrm{diam} \leq D$, $\mathrm{inj} \geq i_0$ (volume comparison), and $V_\Psi \leq V(K, D, i_0, \mu_1)$ from the L^2 non-radiality of the Green kernel (its deviation from the best radial profile, controlled by heat-kernel anisotropy under the same bounds). Hence $\rho_G \geq c(K, D, i_0) > 0$ on the class $\{\iota_\Psi < 18^{-3}\}$—a strictly larger class than the sup-norm near-radial one of Theorem 36, and the one the plateau in the numerical hunt (no family below $\rho_G \approx 0.79$) suggests is broad. Making $V(K, D, i_0, \mu_1)$ explicit is the quantitative frontier item (a).

Theorem 44 (Distributional master inequality, modulus form)**.** *Let Ψ be a monotone radial profile with remainder R, $V_\Psi = \mathbb{E}[R^2] < \infty$, and suppose the pair modulus obeys $\mu_\Psi(\varepsilon) \leq C_1\varepsilon^\theta + C_2$ for some $\theta \in (0, 1]$ and all $\varepsilon > 0$ (the constant slack $C_2 \geq 0$ absorbs range mismatches under perturbation). Then for every $\tau > 0$,*

$$\rho_G(M) \;\geq\; 1 - 3\Big[\, C_1\tau^\theta + \tfrac{8V_\Psi}{\tau^2} \,\Big] - 3C_2,$$

and optimizing at $\tau = (16V_\Psi/(C_1\theta))^{1/(\theta+2)}$,

$$\rho_G(M) \;\geq\; 1 - c(\theta)\, C_1^{2/(\theta+2)}\, V_\Psi^{\theta/(\theta+2)} - 3C_2.$$

For $\theta = 1$ (a Lipschitz modulus, i.e. bounded density with $C_1 = 2\phi_\Psi$, $C_2 = 0$) this recovers Theorem 40, $\rho_G \geq 1 - 18(\phi_\Psi^2 V_\Psi)^{1/3}$, exactly.

Proof. As in Theorem 40, $p \leq \mu_\Psi(\tau) + 2\,\mathbb{P}(|R| \geq \tau/2) \leq C_1\tau^\theta + C_2 + 8V_\Psi/\tau^2$ by Chebyshev, and $\rho_G \geq 1 - 3p$; minimize in τ. □

Remark 45 (Why the modulus form is needed)*.* The bounded-density hypothesis of Theorem 40 ($\phi_\Psi < \infty$) can fail *at* the exact symmetric anchors: on $\mathbb{RP}^d$ the codimension-one cut locus forces $\phi_{\Psi^\star} = \infty$ (Lemma 48 and the admissibility discussion of §17). Since $\mu_\Psi(\varepsilon) \to 0$ always holds by atomlessness (Lemma 4), the modulus form applies where the density form cannot—at the cost of a θ-dependent exponent, the density form being the $\theta = 1$ case. This is the non-radiality-calculus analogue of the (AC)→(AC′) lesson of §7: a bounded density is the wrong hypothesis; the modulus is the right one.

17 A geometric bound: the heat-kernel shell-variance

Purpose. The distributional master inequality (§12, Theorem 40) gives $\rho_G(M) \geq 1-18(\phi_\Psi^2 V_\Psi)^{1/3}$, reducing Green-rank positivity to a bound on the L^2 non-radiality $V_\Psi = \mathbb{E}[(-G - \Psi \circ d_M)^2]$. That Part left V as a quantity to be "bounded by heat-kernel anisotropy" without proof. This Part supplies the reduction. Its three rigorous outputs: (i) the optimal V is exactly the *within-distance-shell variance of the Green kernel*, $V_{\min} = \mathbb{E}[\mathrm{Var}(G \mid d_M)]$; (ii) via the heat representation it equals the integrated *shell-variance of the heat kernel*, which *vanishes identically* *on two-point homogeneous spaces* and (the shell-centered part being $\asymp t^{1-d/4}$, not the crude envelope's $t^{-d/4}$) is finite for $d < 8$—the operative $d \le 3$ coming from the rank-limit step, not the shell-variance; and (iii) an explicit large-time spectral-gap tail plus a quantitative perturbative bound $V_{\min} \le C(g_0)\|g - g_0\|_{C^2}^2$ near any two-point homogeneous g_0. The one non-explicit constant—the small-time curvature-anisotropy amplitude—is isolated and named. Notation follows the preceding Parts; (X, Y) are i.i.d. from the unit-volume measure and all L^2 norms are over $M \times M$ with that measure.

18 The optimal profile is the distance-conditional mean

Proposition 46 (Conditional-variance characterization)**.** *Over all (Borel) radial profiles Ψ, $V_\Psi = \mathbb{E}[(-G(X,Y) - \Psi(d_M(X,Y)))^2]$ is minimized by the distance-conditional mean $\Psi^\star(r) = -\mathbb{E}[G(X,Y) \mid d_M(X,Y) = r]$, and*

$$V_{\min} := V_{\Psi^\star} = \mathbb{E}\big[\mathrm{Var}(G(X,Y) \mid d_M(X,Y))\big].$$

Near the diagonal $\Psi^\star$ is strictly increasing (the $-\gamma_d$ singularity dominates the bounded regular part); its finite anti-concentration $\phi_{\Psi^\star} < \infty$ and its global monotonicity are carried by condition (A*) *below.*

Proof. For fixed r the map $c \mapsto \mathbb{E}[(-G - c)^2 \mid d_M = r]$ is minimized at $c = \mathbb{E}[-G \mid d_M = r]$ with minimal value $\mathrm{Var}(-G \mid d_M = r) = \mathrm{Var}(G \mid d_M = r)$; averaging over the law of $d_M(X,Y)$ gives the claim (the standard L^2-projection onto $\sigma(d_M)$). For the near-diagonal monotonicity: writing $G = \gamma_d \circ d_M + h$ (parametrix lemma, §12) with h bounded, $\Psi^\star = -\gamma_d - \mathbb{E}[h \mid d_M = \cdot]$; for $0 < r_1 < r_2$ small, $\Psi^\star(r_2) - \Psi^\star(r_1) = \big(\gamma_d(r_1) - \gamma_d(r_2)\big) - \big(\mathbb{E}[h \mid r_2] - \mathbb{E}[h \mid r_1]\big)$, and the first term (positive, unbounded as $r_1 \to 0$) dominates the second ($\le 2\|h\|_\infty$) below a threshold, so $\Psi^\star$ is strictly increasing there—using only boundedness of h. The finite anti-concentration $\phi_{\Psi^\star} < \infty$, however, requires control of $\frac{d}{dr}\mathbb{E}[h \mid d_M = r]$, which the parametrix regularity ($h \in C^{0,\alpha}$) does not supply; it is therefore folded into (A*). $\square$

Remark 47 (The admissibility condition (A*)). The bound below controls the *unconstrained* minimizer $V_{\min}$, whose profile $\Psi^\star$ must itself be a monotone profile with finite ϕ before $V_{\min}$ may enter the distributional master inequality. We isolate this as **(A*)** $r \mapsto \mathbb{E}[G \mid d_M = r]$ *is strictly decreasing on* $(0, \operatorname{diam} M)$*, with* $\phi_{\Psi^\star} < \infty$*.* By the Proposition, (A*) holds near the diagonal unconditionally. Its *bounded-density* form, however, can fail even *at* a two-point homogeneous anchor: on the round spheres S^2, S^3 it holds, but on the projective spaces $\mathbb{RP}^2, \mathbb{RP}^3$ the codimension-one cut locus forces $\phi_{\Psi^\star} = \infty$ (Lemma 48), so no C^2-neighborhood can restore it. The right condition is the *modulus* form: $\mu_{\Psi^\star}(\varepsilon) \to 0$ (automatic by atomlessness, Lemma 4), quantified as $\mu_{\Psi^\star}(\varepsilon) \le C_1\varepsilon^\theta + C_2$, which feeds the modulus-form master inequality (Theorem 44) with $\theta = 1$ on the spheres and $\theta = \frac{1}{2}$ on the projective spaces (Lemma 48). It is checkable and measurable (evaluate the concentration function of $\Psi^\star(W)$). We keep the density-form (A*) as the clean $\theta = 1$ statement where it holds, and route the projective and perturbed cases through the modulus form—the non-radiality analogue of the (AC)→(AC$'$) correction of §7.

Lemma 48 (Moduli at the symmetric anchors)**.** *On a two-point homogeneous space $\Psi^\star = -k_0$ exactly, and the flux identity (Paper I) gives $k_0'(r) = -(1 - V(r))/A(r)$ (A the geodesic-sphere area, $V(r) = \operatorname{vol} B_r$), so $f_{\Psi^\star(W)}(s) = f_W(r)/|k_0'(r)| = A(r)^2/(1 - V(r))$ at $s = \Psi^\star(r)$ (near*

$r = 0$ *this is* $\asymp r^{2(d-1)}$*, matching Lemma 34). On* S^d*,* $A \asymp (\pi - r)^{d-1}$ *and* $1 - V \asymp (\pi - r)^d$*, so the density* $\asymp (\pi - r)^{d-2}$ *is bounded for* $d \in \{2, 3\}$ *and* $\mu_{\Psi^\star}(\varepsilon) \le C\varepsilon$ *(*$\theta = 1$*). On* $\mathbb{RP}^d$ *the cut locus has codimension one, so* $A(r) \to A(D) > 0$ *while* $1 - V(r) \asymp A(D)(D - r)$*, giving density* $\asymp (D-r)^{-1} \to \infty$*, i.e.* $\phi_{\Psi^\star} = \infty$*; yet the top-of-range window* $s_{\max} - \Psi^\star(r) = \int_r^D (1 - V)/A \asymp (D - r)^2/2$ *has* W*-mass* $\asymp \sqrt{\varepsilon}$*, so* $\mu_{\Psi^\star}(\varepsilon) \le C\varepsilon + C\sqrt{\varepsilon}$ *(*$\theta = \frac{1}{2}$*).*

19 Heat representation: the shell-variance of the heat kernel

Let $p_t(x, y) = \sum_{k \ge 0} e^{-\mu_k t} \varphi_k(x)\varphi_k(y)$ be the heat kernel ($\mu_0 = 0$, $\varphi_0 \equiv 1$), so $G(x, y) = \int_0^\infty (p_t(x, y) - 1)\, dt$. Write the shell-centering $P_{\mathrm{rad}} F(x, y) = \mathbb{E}[F \mid d_M] = F$'s conditional mean given distance, and $\delta_t = p_t - P_{\mathrm{rad}} p_t$ (the part of the time-t heat kernel that is *not* a function of distance). Define the *time-t heat-kernel non-radiality*

$$\mathcal{N}(t) := \|\delta_t\|_{L^2} = \big(\mathbb{E}[\mathrm{Var}(p_t(X, Y) \mid d_M(X, Y))]\big)^{1/2}.$$

Lemma 49 (Reduction to the heat-kernel non-radiality)**.**

$$V_{\min}^{1/2} \;\le\; \int_0^\infty \mathcal{N}(t)\, dt, \qquad \textit{and} \qquad \mathcal{N}(t) \;\le\; \big(Z(2t) - 1\big)^{1/2},$$

where $Z(s) = \operatorname{Tr} e^{-s\Delta} = \sum_{k \ge 0} e^{-\mu_k s}$ *is the heat trace. Consequently* $\mathcal{N}(t) \equiv 0$ *when* M *is two-point homogeneous (there* p_t *is a function of distance for every* t*), and the envelope integral* $\int_0^\infty (Z(2t) - 1)^{1/2}\, dt$ *converges* iff $d \le 3$.

Proof. The optimal remainder is $R^\star = -G - \Psi^\star \circ d_M = -(G - \mathbb{E}[G \mid d_M]) = -\int_0^\infty (p_t - P_{\mathrm{rad}} p_t)\, dt = -\int_0^\infty \delta_t\, dt$, the interchange of $\int dt$ and the (bounded, linear) conditional expectation being justified by absolute integrability of $p_t - 1$ in L^2 (below). Minkowski's integral inequality gives $V_{\min}^{1/2} = \|R^\star\|_2 \le \int_0^\infty \|\delta_t\|_2\, dt = \int_0^\infty \mathcal{N}(t)\, dt$.

For the envelope: shell-centering is an L^2-orthogonal projection, hence a contraction, so by the law of total variance $\mathcal{N}(t)^2 = \mathbb{E}[\mathrm{Var}(p_t \mid d_M)] \le \mathrm{Var}(p_t(X, Y)) = \mathbb{E}[p_t^2] - 1 = \|p_t\|_2^2 - 1 = \sum_{k \ge 0} e^{-2\mu_k t} - 1 = Z(2t) - 1$ (using $\mathbb{E}[p_t(X, Y)] = \iint p_t = 1$ and Parseval). Two-point homogeneity makes the invariant kernel p_t a function of geodesic distance, so $\delta_t \equiv 0$ and $\mathcal{N} \equiv 0$. Finally the envelope $(Z(2t) - 1)^{1/2}$ is only a crude over-bound on $\mathcal{N}$: as $t \to 0$, $Z(2t) \sim (8\pi t)^{-d/2}$ (Weyl heat-trace asymptotics, vol $= 1$), so $(Z(2t) - 1)^{1/2} \sim c\, t^{-d/4}$; the shell-*centered* $\mathcal{N}(t)$ is strictly smaller, $\asymp t^{1 - d/4}$ near the diagonal (Lemma 53), because the leading MP term is radial and anisotropy enters only at relative order $r^2 \sim t$. Hence $\int_0^\infty \mathcal{N} < \infty$—so $V_{\min} < \infty$, and the interchange above is justified—for all $d < 8$; the tail is exponentially small ($Z(2t) - 1 \le Ce^{-2\mu_1 t}$). ☐

Remark 50 (The genuine role of $d = 4$). The crude envelope integral diverges at $d = 4$, but the shell-variance does not—it is finite for $d < 8$. The operative restriction to $d \le 3$ in this program comes *not* from the shell-variance but from the Hilbert–Schmidt/rank-limit step: the Green-kernel rank-limit theorem (Paper I) requires $\sum_k \mu_k^{-2} < \infty$, i.e. $d < 4$, and §21 restores the chain past it by a fractional filter. Any suggestion that the shell-variance *itself* is finite only for $d \le 3$ is withdrawn: the two thresholds coincide numerically but not by the same mechanism.

20 The explicit large-time tail and the small-time anisotropy

Split $\int_0^\infty \mathcal{N} = \int_0^{t_0} \mathcal{N} + \int_{t_0}^\infty \mathcal{N}$. The tail is fully explicit; the head is reduced to a named curvature-anisotropy amplitude.

Lemma 51 (Explicit spectral-gap tail)**.** *For any $t_0 > 0$, $\displaystyle\int_{t_0}^\infty \mathcal{N}(t)\,dt \le \frac{(Z(2t_0)-1)^{1/2}}{\mu_1}$. Under $\mathrm{Ric} \ge -(d-1)K$ and $\mathrm{diam} \le D$ the spectral gap obeys $\mu_1 \ge \mu_1(K,D) > 0$ (Li–Yau/Zhong–Yang), and $Z(2t_0) - 1 \le C(d,K,D)$ for $t_0 \ge 1$; so the tail is bounded by an explicit function of (d,K,D,t_0) decaying like $e^{-\mu_1 t_0}/\mu_1$.*

Proof. For $t \ge t_0$, $Z(2t)-1 = \sum_{k\ge1} e^{-2\mu_k t} \le e^{-2\mu_1(t-t_0)} \sum_{k\ge1} e^{-2\mu_k t_0} = e^{-2\mu_1(t-t_0)}(Z(2t_0)-1)$, so $\mathcal{N}(t) \le (Z(2t_0)-1)^{1/2} e^{-\mu_1(t-t_0)}$ by Lemma 49; integrating $\int_{t_0}^\infty e^{-\mu_1(t-t_0)}\,dt = 1/\mu_1$ gives the bound. The gap and heat-trace bounds are standard under the stated curvature/diameter hypotheses. □

Definition 52 (Curvature-anisotropy amplitude)**.** *Let $\mathcal{A} := \sup_{0<t\le t_0} t^{d/4-1}\mathcal{N}(t)$, the shell-standard-deviation of the heat kernel on the head interval, normalized by its* shell-centered *small-time scale $t^{1-d/4}$ (Lemma 53)—one power tighter than the crude diagonal envelope $t^{-d/4}$.*

Lemma 53 (Head bound)**.** *For $d < 8$, $\displaystyle\int_0^{t_0} \mathcal{N}(t)\,dt \le \mathcal{A}\int_0^{t_0} t^{1-d/4}\,dt = \frac{4\,\mathcal{A}}{8-d}\,t_0^{(8-d)/4}$. Here $\mathcal{A} < \infty$ always ($\mathcal{N}(t) \asymp t^{1-d/4}$ near 0, so $t^{d/4-1}\mathcal{N}(t)$ is bounded there, and $\mathcal{N}$ is continuous on the compact $(0,t_0]$) and $\mathcal{A} = 0$ on two-point homogeneous spaces (there $\mathcal{N} \equiv 0$). Its* small-time *limit $\lim_{t\to0} t^{d/4-1}\mathcal{N}(t)$ equals the combined shell-oscillation of the Minakshisundaram–Pleijel amplitudes $u_0 \oplus u_1$ (both enter at the same order: the non-radial part of $u_0 = \det(d\exp)^{-1/2}$ is $O(r^2)$·anisotropy with $r^2 \sim t$, and the $u_1 t$ term is $O(1)\cdot t$)—the variation of the Jacobi/curvature data along equal-length geodesics, $O(\sup\|\nabla\,\mathrm{Ric}\|$ and the sectional-curvature variation), vanishing exactly when the space is* two-point homogeneous *(*not *merely locally symmetric: on $S^2\times S^1$, locally symmetric, $\mathrm{Ric}(\dot\gamma,\dot\gamma)$ varies with direction, so u_0 oscillates on shells). The* sup *$\mathcal{A}$, however, is generically dominated by $t \approx t_0$—a spectral (heat-kernel non-radiality) quantity, $\Theta(1)$ for a non-symmetric metric—not by its small-time limit; making it curvature-explicit uniformly in $t \le t_0$ is the frontier item, not claimed here.*

Proof. The integral bound is Definition 52 and $\int_0^{t_0} t^{1-d/4}dt = \frac{4}{8-d}t_0^{(8-d)/4}$ (finite for $d < 8$). For the small-time scaling: near the diagonal the MP expansion $p_t(x,y) = (4\pi t)^{-d/2}e^{-d_M(x,y)^2/4t}\big(u_0(x,y)+u_1(x,y)t+\cdots\big)$ has a *radial* Gaussian factor, so the non-radial part of p_t at leading order is $(4\pi t)^{-d/2}e^{-d_M^2/4t}\big(u_0(x,y) - \mathbb{E}[u_0 \mid d_M]\big)$. Since $u_0 - \mathbb{E}[u_0 \mid d_M] = O(r^2)\cdot$(shell-oscillation), the shell integral $\int (4\pi t)^{-d}e^{-r^2/2t}r^4\,r^{d-1}\,dr \asymp t^{(4-d)/2}$ gives L^2-shell-variance $\mathcal{N}(t)^2 \asymp t^{2-d/2}$ (shell-oscillation of u_0)^2, i.e. $\mathcal{N}(t) \asymp t^{1-d/4}\cdot$(shell-oscillation) and $\lim_{t\to0} t^{d/4-1}\mathcal{N}(t)$ is that oscillation—*not* $t^{d/4}\mathcal{N}(t)$, whose limit is 0. Here $u_0 = \det(d\exp_x(y))^{-1/2}$ is a function of the geometry along the x-to-y geodesic, constant on distance shells exactly when the geometry is isotropic and parallel, and its shell-oscillation is $O(d(x,y)^2)$ times the curvature variation by the Jacobi-field (Van Vleck–Morette) expansion $u_0 = 1 + \frac{1}{12}\mathrm{Ric}(\dot\gamma,\dot\gamma)r^2 + O(r^3)$ (the volume density is $\theta = \sqrt{\det g} = 1 - \frac{1}{6}\mathrm{Ric}(\dot\gamma,\dot\gamma)r^2 + O(r^3)$, and $u_0 = \theta^{-1/2} = (\det g)^{-1/4}$). On a two-point homogeneous space u_0 is a function of distance, so $\mathcal{A} = 0$; the intermediate-time ($t \le t_0$) contribution is likewise 0 there since $\mathcal{N} \equiv 0$. □

Theorem 54 (Geometric L^2 non-radiality bound)**.** *Let M be closed of dimension $d \le 3$,* $\mathrm{vol}(M) = 1$*,* $\mathrm{Ric} \ge -(d-1)K$*,* $\mathrm{diam} \le D$*, injectivity radius $\ge i_0$. Then for every $t_0 \ge 1$,*

$$V_{\min}^{1/2} \;\le\; \frac{4\,\mathcal{A}}{8-d}\,t_0^{(8-d)/4} \;+\; \frac{(Z(2t_0)-1)^{1/2}}{\mu_1},$$

with $\mu_1 \ge \mu_1(K,D) > 0$ and $Z(2t_0) - 1 \le C(d,K,D)$ explicit, and $\mathcal{A}$ the curvature-anisotropy amplitude of Definition 52. Consequently:

(i) $V_{\min} = 0$ on every two-point homogeneous space ($\mathcal{A} = 0$ and $\mathcal{N} \equiv 0$), recovering $\rho_G = 1$ there;

(ii) under (A*) *(Remark 47), combined with the distributional master inequality, $\rho_G(M) \ge 1 - 18\big(\phi_{\Psi^\star}^2 V_{\min}\big)^{1/3}$, so $\rho_G(M) > 0$ whenever the right side of the displayed bound is small enough that $\phi_{\Psi^\star}^2 V_{\min} < 18^{-3}$—a class explicit in (K, D, i_0, μ_1) modulo the single anisotropy amplitude $\mathcal{A}$.*

Proof. Sum Lemmas 53 and 51 into Lemma 49. (i) is $\mathcal{A} = 0$ and $\mathcal{N} \equiv 0$; (ii) is the distributional inequality applied to $\Psi^\star$—in the density form (Theorem 40) where (A*) holds, and in the modulus form (Theorem 44) where $\phi_{\Psi^\star} = \infty$, e.g. the projective anchors (Lemma 48). $\square$

Corollary 55 (Perturbative openness rate, $d \le 3$)**.** *Let g_0 be a two-point homogeneous metric of dimension $d \le 3$* (vol $= 1$) *and g a volume-normalized metric with $\eta := \|g - g_0\|_{C^2} \le \eta_0$ small. Then $V_{\min}(g) \le C(g_0)\,\eta^2$, and*

$$\rho_G(g) \ge 1 - C'(g_0)\,\eta^{2/3} \quad (\textit{near } S^2, S^3), \qquad \rho_G(g) \ge 1 - C'(g_0)\,\eta^{2/5} \quad (\textit{near } \mathbb{RP}^2, \mathbb{RP}^3),$$

both $\to 1$ as $\eta \to 0$; the exponent degrades only at the codimension-one projective (focal) anchors, where the density form of (A*) fails and the modulus form (with $\theta = \frac{1}{2}$) is used. This quantifies the perturbative openness theorem in the L^2 index.

Proof. By Remark 47, (A*) holds for g near g_0, so $V_{\min}(g)$ is admissible. Since g_0 is two-point homogeneous, $\delta_t^{g_0} \equiv 0$, so $\mathcal{N}^g(t) = \|\delta_t^g\|_2 \le \underbrace{\|p_t^g - p_t^{g_0}\|_2}_{\text{(I)}} + \underbrace{\|(I - P_{\mathrm{rad}}^g)p_t^{g_0}\|_2}_{\text{(II)}}$.

(I) parabolic perturbation. Conjugate both heat semigroups to the fixed space $L^2(M, \mathrm{vol}_{g_0})$; the resulting nonnegative self-adjoint operators A_g, A_0 obey $\|A_g - A_0\|_{H^1 \to H^{-1}} \le C\eta$. For $0 < t \le 1$ split Duhamel at $t/2$, placing the Hilbert–Schmidt smoothing on the *long* factor on each half: $\|e^{-(t-s)A_g}(A_g - A_0)e^{-sA_0}\|_{\mathfrak{S}_2} \le C\eta\,(t-s)^{-\frac{1}{2}-\frac{d}{4}}s^{-\frac{1}{2}}$ for $s < \frac{t}{2}$, and symmetrically $C\eta\,(t-s)^{-\frac{1}{2}}s^{-\frac{1}{2}-\frac{d}{4}}$ for $s > \frac{t}{2}$; integrating, $\|e^{-tA_g} - e^{-tA_0}\|_{\mathfrak{S}_2} \le C\eta\,t^{-d/4}$, and the $O(\eta)$ volume-density change in the conjugation is of the same order, so (I)$\le C\eta\,t^{-d/4}$ for $0 < t \le 1$; for $t \ge 1$, on the mean-zero subspace both semigroups contract at rate $e^{-\mu_1(g_0)t/2}$, and the unit-step recursion $d_t \le e^{-c(t-1)}d_1 + e^{-c}d_{t-1}$ with $d_1 \le C\eta$ (the head bound at $t = 1$) gives (I)$\le C\eta\,t\,e^{-c(t-1)}$ (the conjugated ground states differ by $O(\eta)$ and the constant modes cancel after undoing the conjugation).

(II) change of shells. As g_0 is homogeneous, $p_t^{g_0} = q_t \circ d^{g_0}$, and since P_{rad}^g is the best L^2 radial approximant, (II)$\le \|q_t \circ d^{g_0} - q_t \circ d^g\|_2$. The *relative* comparability $|d^g - d^{g_0}| \le C\eta\,d^{g_0}$ (§2, Step 1) with the radial gradient bound $|q_t'(r)| \le Ct^{-(d+1)/2}e^{-r^2/(Ct)}$ and the mean value theorem give, in g_0-polar coordinates, $\|q_t \circ d^{g_0} - q_t \circ d^g\|_2^2 \le C\eta^2 t^{-d-1}\int_0^D r^{d+1}e^{-r^2/(Ct)}dr \le C\eta^2 t^{-d/2}$, i.e. (II)$\le C\eta\,t^{-d/4}$ for $0 < t \le 1$ (and $\le C\eta e^{-ct}$ for $t \ge 1$).

Assemble. $\mathcal{N}^g(t) \le C\eta\, t^{-d/4}$ on $(0,1]$ (integrable since $d \le 3$) and $\le C\eta\, t\, e^{-ct}$ on $[1,\infty)$; hence $V_{\min}(g)^{1/2} = \int_0^\infty \mathcal{N}^g \le C(g_0)\eta$, so $V_{\min}(g) \le C(g_0)\eta^2$. Finally feed $V_{\min}(g)$ to the modulus-form master inequality (Theorem 44) with the transported anchor profile $\Psi = -k_0$ evaluated on d_g (extended linearly past D_{g_0}; the range-mismatch slack is $C_2 = O(\eta)$, lower order), whose modulus is that of Lemma 48 up to the volume-comparable change of the pair law: $\theta = 1$ near S^2, S^3 and $\theta = \frac{1}{2}$ near $\mathbb{RP}^2, \mathbb{RP}^3$. Since the rate exponent is $2\theta/(\theta+2)$ (from $V_{\min} \asymp \eta^2$), this gives $\rho_G(g) \ge 1 - C'(g_0)\eta^{2/3}$ near the spheres and $\ge 1 - C'(g_0)\eta^{2/5}$ near the projective spaces. The projective rate carries one hypothesis we do not prove here: that the C^2 perturbation preserves the *codimension* of the cut locus, so that $1 - V_g$ remains linear at the focal set and the transported modulus stays $\theta = \frac{1}{2}$; this holds for the small perturbations considered (the focal cut locus of a two-point homogeneous space is a nondegenerate submanifold, stable under C^2-small deformation), but a perturbation that degenerated the cut-locus stratum could inflate C_2 and worsen the projective exponent. The exact-anchor conclusion ($\rho_G = 1$ at $\mathbb{RP}^d$, where $V = 0$) is unaffected. □

Remark 56 (Why $t^{-d/4}$, and the limit of this route). The full heat-kernel difference has size $\|p_t^g - p_t^{g_0}\|_2 \asymp \eta\, t^{-d/4}$ —a perturbation of the principal symbol changes the leading Gaussian by a relative $O(\eta)$, with no extra factor of t—which is integrable exactly for $d \le 3$, whence the dimension restriction. The sharper shell-centered scale $\mathcal{N}^g(t) \asymp \eta\, t^{1-d/4}$ (which would extend the bound to $d < 8$) is *not* produced by the triangle split above: that split replaces $(I - P^g_{\mathrm{rad}})p^g_t$ by a full difference plus a shell change and so destroys the radial cancellation; obtaining it would require a direct estimate of $(I - P^g_{\mathrm{rad}})p^g_t$. Since the Green-rank chain operates at $d \le 3$, the $t^{-d/4}$ envelope suffices.

21 Past the critical dimension: the fractional filter

Purpose. The rank form of Angular Preservation is settled below the critical dimension $d = 4$: the commute angular ranking converges uniformly in the mode count to the Green-kernel ranking (Paper I, Green-kernel rank-limit theorem), reducing the question to Green-rank positivity. At $d = 4$ the commute Green kernel $G = \sum_k \mu_k^{-1} \varphi_k \otimes \varphi_k$ ceases to be Hilbert–Schmidt ($\sum_k \mu_k^{-2} = \infty$ by Weyl's law $\mu_k \asymp k^{2/d}$), so the limit step fails. Paper I flags the remedy—a stronger filter $\mu_k^{-s/2}$, $s > d/4$—and the first question, monotonicity of the fractional Green kernel on two-point homogeneous spaces. This Part carries the entire chain up one level to the fractional filter and, using heat-kernel subordination, proves the two-point homogeneous case *in every dimension*, so that the reduction “Angular Preservation ⇐ fractional-Green-rank positivity” holds for all d, with the geometric coefficient equal to 1 on the rank-one symmetric spaces regardless of dimension. Notation follows Paper I and §12.

22 The fractional filter and Hilbert–Schmidt stability

Fix $s > 0$ and, at a multiplet-closed cutoff Λ, define the s-filtered embedding and its kernel

$$F_{s,\Lambda} = \big(\mu_k^{-s/2} \varphi_k\big)_{0<\mu_k\le\Lambda}, \quad K_{s,\Lambda}(x,y) = \sum_{0<\mu_k\le\Lambda} \frac{\varphi_k(x)\varphi_k(y)}{\mu_k^s}, \quad S_{s,\Lambda}(x) = K_{s,\Lambda}(x,x), \quad N_{s,\Lambda} = \frac{F_{s,\Lambda}}{\sqrt{S_{s,\Lambda}}}.$$

The commute filter is $s = 1$. Write $K_s = \sum_{k\ge1} \mu_k^{-s} \varphi_k \otimes \varphi_k$ for the full s-kernel (the kernel of Δ^{-s} on the zero-mean subspace) and $\rho_{G,s}(M) := \rho_S\big(d_M(X,Y), -K_s(X,Y)\big)$ for the *fractional*

Green-rank coefficient.

Lemma 57 (Hilbert–Schmidt stability)**.** $\|K_{s,\Lambda'} - K_{s,\Lambda}\|^2_{L^2(M\times M)} = \sum_{\Lambda<\mu_k\le\Lambda'} \mu_k^{-2s}$*, and by Weyl's law* $\mu_k \asymp k^{2/d}$ *this converges iff* $s > d/4$*. Hence for* $s > d/4$*,* $K_{s,\Lambda} \to K_s$ *in* $L^2(M \times M)$*, and* $K_s \in L^2$*.*

Proof. The $\varphi_k \otimes \varphi_k$ are orthonormal in $L^2(M\times M)$, giving the stated norm; $\sum_k \mu_k^{-2s} \asymp \sum_k k^{-4s/d}$ converges iff $4s/d > 1$. □

For $d \ge 4$ any $s > d/4 > 1$ works; the minimal admissible filter grows with dimension.

23 The fractional rank limit

Theorem 58 (Fractional Green-kernel rank limit)**.** *Let* M *be closed of dimension* d*,* $\mathrm{vol}(M) = 1$*, and* $s > d/4$*. Let* (X, Y) *have bounded density with respect to* $\mathrm{vol} \otimes \mathrm{vol}$*, with the laws of* $d_M(X,Y)$ *and* $K_s(X,Y)$ *atomless. Then*

$$\rho_S\big(d_M(X,Y), \|N_{s,\Lambda}(X) - N_{s,\Lambda}(Y)\|\big) \;\longrightarrow\; \rho_S\big(d_M(X,Y), -K_s(X,Y)\big) = \rho_{G,s}(M),$$

and likewise for Kendall's τ*. Consequently, if* $\rho_{G,s}(M) > 0$ *there are* $\Lambda_0 < \infty, \rho_0 > 0$ *with the* s*-filtered angular ranking* $\ge \rho_0$ *for every multiplet-closed* $\Lambda \ge \Lambda_0$*.*

Proof. Identical in structure to the $s = 1$ theorem (Paper I, Green-kernel rank-limit), with μ_k^{-1} replaced by μ_k^{-s} throughout. *(1)* Hilbert–Schmidt limit is Lemma 57. *(2)* The diagonal $S_{s,\Lambda}(x) = \sum_{\mu_k\le\Lambda} \mu_k^{-s}\varphi_k(x)^2$ is asymptotically constant uniformly in x: the uniform pointwise Weyl law $\sum_{\mu_k\le L} \varphi_k(x)^2 = c_d L^{d/2} + O(L^{(d-1)/2})$ (Hörmander, uniform in x) and partial summation give, for $d/4 < s \le d/2$, $S_{s,\Lambda}(x) = A_{s,\Lambda} + o(A_{s,\Lambda})$ uniformly in x, where $A_{s,\Lambda}$ is the leading term of $\int^\Lambda L^{-s}\, dc_d L^{d/2}$: $\asymp \Lambda^{d/2-s}$ (diverges) for $s < d/2$, $\asymp \log\Lambda$ for $s = d/2$ (the uniform Hörmander remainder $O(L^{(d-1)/2})$ contributes a relative $O(\Lambda^{-1/2})$, lower order). For $s > d/2$ the diagonal instead *converges* to a finite, generally non-constant $S_s(x)$ and does not rescale to a constant—this case is the Remark. *(3)* Rescale $H_{s,\Lambda} = A_{s,\Lambda} C_{s,\Lambda} \to K_s$ in L^2, ranks unchanged by the positive constant. *(4)* Bounded pair density carries L^2-convergence to convergence in probability; atomlessness of the limit gives distribution-function convergence and hence convergence of the population Spearman/Kendall coefficients. □

Remark 59 (The convergent-diagonal regime $s > d/2$)*.* When $s > d/2$ the diagonal $S_{s,\Lambda}(x) \to S_s(x) = \sum_k \mu_k^{-s}\varphi_k(x)^2$, a *finite, generally non-constant* function of x (the fractional analogue of a local density). Step (2) then does not rescale to a constant; instead $C_{s,\Lambda} \to K_s(x,y)/\sqrt{S_s(x)S_s(y)}$ pointwise and in L^2, and the rank limit is $\rho_S(d_M, -K_s/\sqrt{S_s S_s})$. The normalization by S_s is exactly the row-normalization of the s-embedding; on a homogeneous space S_s is constant and this reduces to $\rho_{G,s}$. For the inhomogeneous case the operative coefficient is the S_s*-normalized* fractional Green-rank correlation, and Theorem 58 holds with K_s replaced by $K_s/\sqrt{S_s(x)S_s(y)}$; all subsequent statements carry the same replacement. We keep K_s in the display for the homogeneous and $d/4 < s \le d/2$ cases, where $S_{s,\Lambda}$ genuinely rescales to a constant.

24 The fractional master inequality and the Riesz parametrix

The master inequality is filter-agnostic: it used only that $-K$ is compared to a monotone radial profile. Restating (§12, Theorem 32): for any strictly increasing Ψ with remainder $R_s = -K_s - \Psi \circ d_M$,

$$\rho_{G,s}(M) \;\geq\; 1 - 3\,\mu_\Psi\big(2\|R_s\|_\infty\big) \;\geq\; 1 - 12\,\phi_\Psi \|R_s\|_\infty,$$

with $\rho_{G,s} = 1$ when $R_s \equiv 0$. The near-diagonal remainder is again bounded, now by the Riesz parametrix.

Lemma 60 (Riesz parametrix)**.** *Let M be closed smooth,* $\mathrm{vol}(M) = 1$*, and $s > (d-1)/2$ (a nonempty range for every d, and automatically $> d/4$ since $d > 2$). Near the diagonal $K_s(x, y) = \gamma_{d,s}\, d_M(x, y)^{2s-d} + h_s(x, y)$ with $\gamma_{d,s} > 0$ the universal Riesz constant and h_s bounded up to the diagonal; the leading profile $r \mapsto \gamma_{d,s} r^{2s-d}$ (exponent $2s-d \in (-1, 0)$, strictly decreasing) is radial with a point-independent coefficient, and it is the* only *singular part—the subleading Hadamard corrections carry exponent $2s - d + 2j \geq 2s - d + 2 > 1$ for $j \geq 1$, hence are bounded even though their coefficients are point-dependent. When $2s \geq d$, K_s is bounded (continuous, $2s > d$) or $-c \log d_M +$bounded ($2s = d$). In all cases $\Psi_s = -\gamma_{d,s} r^{2s-d}$ (resp. the bounded/log profile), extended strictly increasing on $[r_0, \mathrm{diam}\, M]$, has finite $\|R_s\|_\infty$ and finite ϕ_{Ψ_s}.*

Proof. $K_s = \Delta^{-s}$ on the zero-mean subspace is the Riesz potential of order $2s$; it is a classical pseudodifferential operator of order $-2s$ with principal symbol $|\xi|_g^{-2s}$, and its kernel has the Hadamard expansion $K_s(x, y) = \sum_{j\geq 0} c_j(x, y) d_M(x, y)^{2s-d+2j} +$ (smooth) with $c_0 = \gamma_{d,s}$ the universal Euclidean Riesz constant (the metric is Euclidean to second order in normal coordinates) and c_j smooth for $j \geq 1$. The restriction $s > (d - 1)/2$ makes $2s - d > -1$, so every $j \geq 1$ term has exponent $2s - d + 2j > 1$ and is bounded up to the diagonal (with bounded, point-dependent coefficient); thus only $c_0 d_M^{2s-d}$ is singular, and $h_s := K_s - \gamma_{d,s} d_M^{2s-d}$ is bounded. Hence $R_s = -K_s - \Psi_s \circ d_M = -h_s$ near the diagonal, bounded; off the diagonal both K_s and $\Psi_s \circ d_M$ are bounded. Finiteness of ϕ_{Ψ_s} is the change-of-variables computation of §12 with exponent $2s - d$ in place of $2 - d$: $\Psi_s' \asymp r^{2s-d-1}$ and $f_W \asymp r^{d-1}$ give density $\asymp r^{2(d-s)}$, which $\to 0$ at $r \to 0$ for $s < d$, and is bounded on $[r_0, \mathrm{diam}\, M]$. ☐

Theorem 61 (Fractional near-radial positivity)**.** *Let $s > (d - 1)/2$. With Ψ_s the Riesz profile of Lemma 60 (a* single *fixed profile, both ϕ_{Ψ_s} and $\|R_s\|_\infty$ finite), $\rho_{G,s}(M) \geq 1 - 12\,\phi_{\Psi_s} \|R_s\|_\infty$, so $\rho_{G,s}(M) > 0$ whenever $\phi_{\Psi_s} \|R_s\|_\infty < 1/12$; and by Theorem 58 the s-filtered angular ranking is then uniform-in-mode-count rank-faithful. The perturbative openness argument (§2) transfers with $\mu_k^{-1} \to \mu_k^{-s}$—its spectral-tail step now requires $\sum_k \mu_k^{-2s} < \infty$ (i.e. $s > d/4$), the resolvent contour weight becomes z^{-s} (holomorphic off the negative axis, with 0 excluded), and atomlessness of the K_s-pair law is assumed as in Theorem 58—so $\rho_{G,s} > 0$ on a C^2-open set of metrics containing the exact cases of §4.*

25 Two-point homogeneous spaces, all dimensions, via subordination

The one place the fractional story is *cleaner* than the commute case is the exact computation: there is no local Laplace ODE for Δ^{-s}, but subordination to the heat semigroup supplies strict monotonicity in every dimension at once.

Theorem 62 (Fractional Green-rank $=1$ on rank-one symmetric spaces)**.** *Let M be a compact two-point homogeneous space (a sphere S^d, or a projective space $\mathbb{RP}^d, \mathbb{CP}^d, \mathbb{HP}^d, \mathbb{OP}^2$) of any dimension,* $\mathrm{vol}(M) = 1$*, and let $s > 0$. Then $K_s(x, y) = k_s(d_M(x, y))$ with k_s strictly decreasing on* $(0, \operatorname{diam} M)$*; hence $\rho_{G,s}(M) = 1$, and the S_s-normalized coefficient of the Remark is likewise* 1*. In particular, for every d and every admissible $s > d/4$, Angular Preservation (fractional-filter rank form) holds on these spaces, uniformly in the mode count.*

Proof. By subordination, for $s > 0$ and $\lambda > 0$, $\lambda^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda t}\, dt$; applied to Δ on the zero-mean subspace,

$$K_s(x, y) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \big(p_t(x, y) - 1\big)\, dt,$$

where p_t is the heat kernel and the -1 removes the constant mode ($\mathrm{vol}\, M = 1$). Two-point homogeneity makes p_t a function of geodesic distance, $p_t(x, y) = P_t(d_M(x, y))$. The genuinely sharp radial-derivative estimate of Nowak–Sjögren–Szarek [15] on compact rank-one symmetric spaces gives

$$-\partial_r P_t(r) \asymp r\,(D - r)\, H_t(r), \qquad 0 \le r \le D := \operatorname{diam} M, \quad t > 0,$$

with $H_t(r) > 0$ for $0 < r < D$; in particular $\partial_r P_t(r) < 0$ on $(0, D)$ for every $t > 0$, and $\partial_r P_t(0) = \partial_r P_t(D) = 0$—so the pole, the antipodal focal boundary, and the singular radial drift are all subsumed in the cited estimate. Rather than differentiate K_s under the integral, fix $0 < r_1 < r_2 < D$: strict radial monotonicity gives $P_t(r_1) - P_t(r_2) > 0$ for every $t > 0$, so

$$k_s(r_1) - k_s(r_2) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \big(P_t(r_1) - P_t(r_2)\big)\, dt \;>\; 0,$$

the integral finite by off-diagonal Gaussian decay at $t \to 0$ and exponential relaxation to the constant mode at $t \to \infty$. Thus k_s is strictly decreasing on $(0, D)$, $-K_s$ is a strictly increasing function of d_M, and—the pair-distance law being atomless—the two carry identical population ranks, so $\rho_{G,s} = 1$. In the convergent-diagonal regime $s > d/2$, homogeneity makes $S_s(x) = K_s(x, x)$ a positive constant, and division by it does not change ranks. □

Remark 63 (Provenance of the analytic input). The one analytic ingredient—strict radial monotonicity of the heat kernel with the vanishing of its radial derivative at both endpoints—is imported from the published sharp estimate [15] rather than argued by an ad hoc maximum principle; this is deliberately safer, as a self-contained treatment must control the parametrix uniformly up to the antipodal focal set, which is delicate. (The citation has been verified against the source: [15], Corollary 4.2, gives the sharp radial-derivative bounds and states explicitly that the heat kernel is strictly decreasing in geodesic distance on every compact rank-one symmetric space.)

Remark 64 (What this settles). Theorem 62 removes the dimension ceiling from the exact case: the fractional Green-rank coefficient is 1 on all rank-one symmetric spaces in *every* dimension, for every $s > 0$—strictly stronger than the $s = 1$, $d \le 3$ corollary of Paper I, and proved by a different (subordination) route that never needs Hilbert–Schmidt stability. Combined with Theorem 58 ($s > d/4$) and Theorem 61 (openness), the reduction of Angular Preservation to Green-rank positivity now holds in all dimensions with the fractional filter, and its exact anchor—value 1 on the symmetric spaces—is dimension-free. The residual open problem is the same shape as before, one level up: fractional-Green-rank positivity $\rho_{G,s}(M) > 0$ for general M, with the near-radial class of Theorem 61 as the proven interior.

26 Standing facts, the frontier, and open problems

Standing facts. The proofs draw on the following standard tools, cited at point of use: the level-set/coarea lemmas and Łojasiewicz sublevel estimates ([10, 24, 25, 26, 23]); cut-locus and heat-kernel regularity ([11, 19]; the Green-function parametrix, [17, 21, 20]); elliptic regularity and analyticity of solutions ([18, 22]); norm resolvent convergence of uniformly comparable forms ([12]); the uniform pointwise Weyl law ([13]); the spectral-gap lower bound under a Ricci lower bound and diameter ([14, 16]); Daniels' inequality, the Diaconis–Graham footrule bound, and U-statistic consistency for grade correlations ([27, 29, 28]); and the Duhamel heat-kernel perturbation estimate. The spectral-embedding background—the commute-time collapse, row-normalization, diffusion maps, and truncated-eigenmap embeddings—is [2, 3, 4, 7, 8, 9], with the graph-Laplacian convergence rates of [5, 6]. None is load-bearing beyond its stated role.

What is now reduced to what. For $2 \leq d \leq 3$, Angular Preservation (rank form) is implied by: (i) the rank transfer, which at a *fixed* cutoff needs calibration $E_n \to 0$ and (AC$'$)—proved for real-analytic metrics (Part 7) and, more generally, for smooth metrics under the spectral nondegeneracy (ND) (Part 11)—and which at a *growing* cutoff ($E_n\sqrt{A_m} \to 0$) needs *only* calibration, (AC$'$) being discharged; and (ii) Green-rank positivity, reduced to a single scale-invariant non-radiality index (Part 12), itself the heat-kernel shell-variance, which vanishes on two-point homogeneous spaces and is controlled by an explicit spectral-gap tail and one curvature-anisotropy amplitude (Part 17). Consequently the *growing-window* rank form is not merely implied by but *equivalent* to $\rho_G(M) > 0$ (Part 11). Hypotheses (H3) and atomlessness are deleted outright (Part 2); the critical dimension is crossed by the fractional filter (Part 21).

A numerical counterexample hunt. Green-rank positivity may be false on extreme geometries; the master inequality gives any counterexample the exact target $\rho_G \leq 0$. A registered numerical hunt (public repository, `experiments/green-rank`) computes ρ_G across four geometry families—deformed flat tori, thin-neck surfaces of revolution, Berger spheres, and topological handles—each with a control validating the discrete pipeline. None yields a counterexample: ρ_G never drops below ≈ 0.79, and the two candidates the theory flags (thin necks and Berger spheres) both clear. The homogeneous families rise to $\rho_G = 1$; inhomogeneity lowers ρ_G but plateaus, consistent with the shell-variance staying bounded. This is accumulating evidence, not proof.

Open, in order of expected difficulty. (a) Remove (ND): decide whether the spectral nondegeneracy of Part 11—which already yields fixed-cutoff (AC$'$) for all smooth metrics—can fail for any smooth metric. The sole obstruction is a positive-measure set on which $(\Delta - \mu_j)\sqrt{S_\Lambda}$ is infinitely Δ-flat at the exact eigenvalue μ_j; it is absent in the analytic category. (b) Make the curvature-anisotropy amplitude of Part 17 explicit in $(K, D, i_0, \sup\|\nabla \operatorname{Ric}\|)$, promoting the leading-order estimate to a bound uniform in the head time-interval; this converts the named-$\mathcal{A}$ class into a closed curvature/diameter theorem. (c) Green-rank positivity beyond the perturbative and near-radial regimes—whether the plateau observed numerically reflects a universal lower bound $\rho_G \geq c > 0$ for surfaces, a statement strictly stronger than positivity. (d) The general $d \geq 4$ positivity for the fractional filter—the same question one level up, with the exact anchor already in hand.

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